On de Bruijn Array Codes—Part II: Pseudo-Random Array Codes

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***Abstract*—A pseudo-random array is a two-dimensional array in which each $n_1 \times n_2$ nonzero matrix is contained exactly once as a window in the array. The pseudo-random arrays we consider have many desirable properties in addition, such as the shift-and-add-property, i.e., the addition of the array to any of its nontrivial shifts is another nontrivial shift of the array. A pseudo-random array code is a linear code of $r_1 \times r_2$ arrays in which each $n_1 \times n_2$ nonzero matrix is contained exactly once as a window in one of the arrays. In this paper, new parameters for pseudo-random arrays are presented, and their construction is generalized to pseudo-random array codes. Our constructions of pseudo-random array codes are based on the folding of sequences. Two techniques to verify whether an array or a set of arrays constructed by folding is a pseudo-random array or a pseudo-random array code, respectively, are presented. These verification techniques can also be used for VLSI testing.**

***Index Terms*—de Bruijn arrays, folding, irreducible polynomials, pseudo-random array codes, shift-register sequences.**

I. Introduction

GENERALIZATIONS of one-dimensional sequences and codes to higher dimensions are quite natural from both theoretical and practical points of view. Such generalizations have been considered for various structures such as error-correcting codes [2], [30], burst-correcting codes [3], [4], [11], [12], constrained codes [34], [39], and de Bruijn sequences [6], [24], [28]. This paper considers a generalization of one-dimensional linear sequences with a window property to two-dimensional linear arrays with a window property. For simplicity, only binary arrays and binary sequences are considered in this paper. However, some of the results can be generalized to any alphabet of a finite field $\mathbb{F}_q$ with q elements, as was done in other papers [22], [26].

Received 19 August 2025; revised 1 April 2026; accepted 13 April 2026. Date of publication 21 April 2026; date of current version 16 July 2026. The work of Tuvi Etzion was supported in part by Israel Science Foundation under Grant 222/19. An earlier version of this paper was presented in part at the IEEE International Symposium on Information Theory 2024 [DOI: 10.1109/ISIT57864.2024.10619645] and in part at the IEEE International Symposium on Information Theory 2025 [DOI: 10.1109/ISIT63088.2025.11195674]. *(Corresponding author: Tuvi Etzion.)*

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Communicated by C. Carlet, Associate Editor for Sequences and Cryptography.

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TIT.2026.3686267.

Digital Object Identifier 10.1109/TIT.2026.3686267

A. Definitions of Arrays and Codes

A ***span n de Bruijn sequence*** is a cyclic sequence of length 2^n in which each n-tuple is contained in exactly one window of n consecutive digits. A ***span n shortened de Bruijn sequence*** is a sequence of length $2^n - 1$ in which each nonzero n-tuple is contained in exactly one window of n consecutive digits. A ***span n M-sequence*** (pseudo-random sequence) $\mathcal{S}$ is a span n shortened de Bruijn sequence such that if $\mathcal{S}'$ is a nontrivial shift of $\mathcal{S}$, then $\mathcal{S} + \mathcal{S}'$ is another nontrivial shift of $\mathcal{S}$. This is the ***shift-and-add property*** of an M-sequence. The following definition generalizes the definition of a de Bruijn sequence to a two-dimensional array.

Definition 1: A ***de Bruijn array*** (often known as a ***perfect map***) is an $r_1 \times r_2$ doubly-periodic array (cyclic horizontally and vertically like in a torus), such that each $n_1 \times n_2$ binary matrix appears exactly once as a window in the array.

Perfect maps were first presented by Reed and Stewart [29] and were considered in many papers that discussed constructions and applications of these arrays. The definition was generalized in [9] and [10] to a set of arrays, i.e., a code which contains arrays of the same dimensions, as follows.

Definition 2: A ***de Bruijn array code*** is a set of $r_1 \times r_2$ doubly-periodic arrays, such that each $n_1 \times n_2$ matrix appears exactly once as a window in one of the arrays. Such a set of arrays will be called an $(r_1, r_2; n_1, n_2)$-DBAC.

In general, the ***size*** Δ of an array code is the number of arrays (codewords) in the code. In the previous part of this work [10], several direct and recursive constructions for de Bruijn array codes are presented.

Definition 3: A ***shortened de Bruijn array*** (or a ***shortened perfect map***) is an $r_1 \times r_2$ doubly-periodic array, such that $r_1 r_2 = 2^{n_1 n_2} - 1$ and each nonzero $n_1 \times n_2$ matrix appears exactly once as a window in the array. Such an array will be called an $(r_1, r_2; n_1, n_2)$-SDBA. We note that some authors call an SDBA a pseudo-random array, but in this paper we require pseudo-random arrays to have an extra property; see below. A ***shortened de Bruijn array code*** $\mathbb{C}$ is a set of $r_1 \times r_2$ arrays such that $r_1 r_2$ divides $2^{n_1 n_2} - 1$ and each nonzero $n_1 \times n_2$ matrix is contained exactly once in one of the arrays. Such a code will be referred to as an $(r_1, r_2; n_1, n_2)$-SDBAC.

The following lemma is an immediate consequence of the definition.

Lemma 1: If $\mathbb{C}$ is an $(r_1, r_2; n_1, n_2)$-SDBAC of size Δ, then

1. $r_1 > n_1$ or $r_1 = n_1 = 1$,
2. $r_2 > n_2$ or $r_2 = n_2 = 1$, and
3. $\Delta r_1 r_2 = 2^{n_1 n_2} - 1$.

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Definition 4: An $(r_1, r_2; n_1, n_2)$ ***pseudo-random array*** (later referred to as $(r_1, r_2; n_1, n_2)$-PRA for short) $\mathcal{A}$ is an $(r_1, r_2; n_1, n_2)$-SDBA, for which if $\mathcal{A}'$ is a nontrivial shift (both horizontal and vertical shifts, where at least one of them is not trivial) of $\mathcal{A}$, then $\mathcal{A} + \mathcal{A}'$ is also a shift of $\mathcal{A}$, where the addition is performed bit-by-bit. This is the ***shift-and-add property*** of the PRA.

Pseudo-random arrays were constructed first in [22], [26], and [36] and they (along with perfect maps) have found various important applications. They were used in pattern recognition for structured light systems as described in Geng [13], Lin et al. [21], Morano et al. [25], Salvi et al. [32], and Salvi et al. [33]. They are also used in transforming a planar surface into a sensitive touch screen display, see Dai and Chung [5], in camera localization as described by Szentandrasi et al. [38], and in one-shot shape acquisition, e.g., Pagès et al. [27]. Finally, they can be applied to surface measurements as described in Kiyasu et al. [18] and in Spoelder et al. [37], and also in coded aperture imaging as was presented, for example, in Gottesman and Fenimore [16].

Definition 5: An $(r_1, r_2; n_1, n_2)$ ***pseudo-random array code*** ($(r_1, r_2; n_1, n_2)$-PRAC for short) $\mathbb{C}$ is an $(r_1, r_2; n_1, n_2)$-SDBAC for which if $\mathcal{A}, \mathcal{B} \in \mathbb{C}$ ($\mathcal{A}$ and $\mathcal{B}$ are either distinct or $\mathcal{B}$ is a nontrivial shift of $\mathcal{A}$), then $\mathcal{A} + \mathcal{B}$ is also a codeword of $\mathbb{C}$. This is the ***shift-and-add property*** of the pseudo-random array code.

The shift-and-add property of a pseudo-random array code implies that its codewords and all their possible cyclic shifts horizontally and vertically together with the $r_1 \times r_2$ all-zero matrix, form a linear array code.

B. Contributions and Outline

In the current work, constructions for pseudo-random arrays and pseudo-random array codes are presented. The codes are formed by folding nonzero sequences generated by a polynomial whose nonzero sequences have the same length. This technique is well known, but it is not known which folded sequences yield such arrays and array codes. Two proof techniques are used to prove the correctness of the suggested construction methods. The first is based on the linear span of the bitwise product of sequences generated by two polynomials, where the non-zero sequences produced by each polynomial must all have the same length. The second proof technique is a direct generalization of a proof method suggested by MacWilliams and Sloane [22]. This proof technique gives pseudo-random arrays with new parameters which were not obtained in [22], and many pseudo-random array codes which cannot be obtained using the first proof technique. Many codes can be obtained using both proof techniques; however, each technique yields codes that are not obtained by the other technique. Two necessary and sufficient conditions to verify whether folding yields a PRA or a PRAC are also presented. These conditions can also be used for VLSI testing. The main results of this work are presented in Theorems 2, 3, 4, 5, and 6 and their consequences.

The rest of this paper is organized as follows. In Section II, we present the basic definitions and theory for the arrays considered here. Section III introduces the main technique in this paper, known as folding. This technique is used to form a PRA, and a proof for PRAs with new parameters is given in this section (Theorem 2). Section IV generalizes the folding technique so as to construct PRACs. Surprisingly, a different method is used to generate some of these codes, which suggests another way to prove the correctness of the construction. However, codes with new parameters are also introduced by this method. Section V presents a necessary and sufficient condition for the folding technique to produce a PRA or a PRAC (Theorem 3). The method presented is applied only to sequences obtained from irreducible polynomials, and it can also be used for VLSI testing. Section VI introduces the first proof technique for the constructed array codes (Theorem 4). Section VII presents a generalization of the proof technique used in Section III to prove the existence of more PRACs based on folding (Theorem 5). Folding can be used with sequences of various types of polynomials (primitive, irreducible and not primitive, and reducible) for which the generated nonzero sequences have the same length. Analysis of the parameters of PRACs obtained by the various constructions is given in Section VIII. This section also suggests hierarchies between the various codes and a comparison between the array codes obtained by the two proof techniques. Section IX contains another necessary and sufficient condition to verify whether the array or code obtained from folding is a PRA or a PRAC, respectively (Theorem 6). A conclusion and discussion of further research is given in Section X.

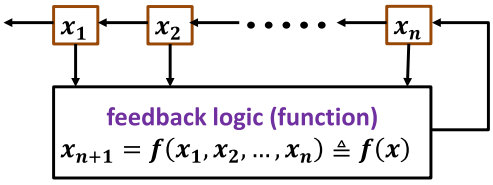


Fig. 1. Feedback shift register of order n.

II. Preliminaries

We start this section with the theory of shift-register sequences. Shift registers and their sequences have been extensively studied, and their theory can be found in the following comprehensive books: [8], [14], [15], [20]. The material in this section is presented in these books. This theory will be applied later to our two-dimensional arrays.

A. Shift Registers

A ***feedback shift register*** of order n (an FSR_n in short) has 2^n ***states***, represented by the set of 2^n binary words of length n. An FSR_n has n cells, which are binary storage elements, where each cell stores one of the bits of the current state $x = (x_1, x_2, \ldots, x_n)$. Such an FSR_n is depicted in Fig. 1.

If the word $(x_1, x_2, \ldots, x_n)$ is the state of the FSR_n, then x_i is stored in the i-th cell of the FSR_n. The n cells are connected to another logic element which computes a Boolean function $f(x_1, x_2, \ldots, x_n)$. At periodic intervals, controlled and synchronized by a global clock, x_2 is transferred to x_1, x_3 to x_2, and so on until x_n is transferred to x_{n-1}. The value of the feedback function is transferred to x_n. Hence, it is common to write $x_{n+1} = f(x_1, x_2, \ldots, x_n)$. The register starts with an

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initial state $(a_1, a_2, \ldots, a_n)$, where a_i, $1 \le i \le n$, is the initial value stored in the i-th cell.

A ***linear feedback shift register*** of order n (LFSR$_n$ in short) is a shift register whose feedback function is linear. We are interested only in nonsingular LFSRs. Such an LFSR$_n$ has a feedback function of the form

$$x_{n+1} = f(x) = f(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n} c_i x_{n+1-i},$$
$$c_i \in \{0, 1\},\ 1 \le i \le n-1,\ c_n = 1.$$

The associated LFSR$_n$ sequence $(a_k)_{k=-n}^{\infty}$, where the initial state is $(a_{-n}, a_{-n+1}, \ldots, a_{-1})$, satisfies a linear recursion

$$a_k = \sum_{i=1}^{n} c_i a_{k-i}, \quad k = 0, 1, \ldots. \tag{1}$$

The ***characteristic polynomial*** of the sequence (a_k) is defined by

$$c(x) = 1 - \sum_{i=1}^{n} c_i x^i.$$

We say that the (characteristic) polynomial $c(x)$ ***generates*** the (infinite) sequence (a_k). Any such polynomial can be written in a compact way as $c_n c_{n-1} \cdots c_1 1$. We sometimes think of different shifts of the sequence (a_k) as the same (i.e. we think of the sequence as a cyclic sequence; see below). The same sequence might be generated by more than one LFSR, and so might be represented in several different ways.

B. Sequences

The sequences generated by various characteristic polynomials are used here to form the PRACs. Hence, we will define a few properties of these sequences and a few operations on them. These properties and operations will be used to construct PRACs or to prove the correctness of the constructions.

We distinguish between acyclic and cyclic sequences. A ***cyclic sequence*** is denoted by $[a_0 a_1 \cdots a_{n-1}]$ and an ***acyclic sequence*** (a word) is denoted by $(a_0 a_1 \cdots a_{n-1})$. An acyclic sequence has a starting position. In contrast, we think of different shifts of a cyclic sequence as being equal. Cyclic sequences can sometimes be viewed as acyclic sequences. For example, when such sequences are concatenated, the sequences are considered as acyclic sequences, but the outcome might be a cyclic sequence. The ***length*** of a sequence $\boldsymbol{a} = [a_0, a_1, \cdots, a_{n-1}]$ is n. The length of the sequence $\boldsymbol{a}$ will be denoted by $\ell(\boldsymbol{a})$. The ***least period*** of a sequence $\boldsymbol{a} = [a_0, a_1, \cdots, a_{n-1}]$ is the least positive integer π such that $a_i = a_{i+\pi}$ for all $0 \le i \le n-1$, where subscripts are computed modulo n. The least period of the sequence $\boldsymbol{a}$ will be denoted by $\pi(\boldsymbol{a})$. An integer p such that $a_i = a_{i+p}$ for all $0 \le i \le n-1$ is a ***period*** of the sequence. This implies that if a sequence $\boldsymbol{a}$ has length $k\pi$ and $\pi = \pi(\boldsymbol{a})$, then $\kappa\pi$ is a period of $\boldsymbol{a}$ if and only if κ divides k.

Similar definitions are adopted and given for arrays. In particular, the ***horizontal period*** of an $r_1 \times r_2$ array $\mathcal{A}$ is the least integer π such that π is a period of each row of $\mathcal{A}$. Similarly, the ***vertical period*** of an $r_1 \times r_2$ array $\mathcal{A}$ is the least integer π such that π is a period of each column of $\mathcal{A}$. In all the $r_1 \times r_2$ arrays considered here, the horizontal period will be r_2 and the vertical period will be r_1.

Let $\boldsymbol{a} = a_0, a_1, \ldots$ and $\boldsymbol{b} = b_0, b_1, \ldots$ be two sequences (either cyclic or acyclic) of the same length. the ***(bitwise) addition*** of $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as $\boldsymbol{a}+\boldsymbol{b} = a_0+b_0, a_1+b_1, \ldots$ and the ***bitwise product*** of $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as $\boldsymbol{a} \cdot \boldsymbol{b} = a_0 b_0, a_1 b_1, \ldots$. Furthermore, the bitwise product of two matrices $A = (a_{i,j})$ and $B = (b_{i,j})$ in $\mathbb{F}_2^{r_1 \times r_2}$ is defined as $A \cdot B = (a_{i,j} b_{i,j})$, where $0 \le i \le r-1$ and $0 \le j \le t-1$. Similarly, the (bitwise) addition of two matrices is defined.

For two positive integers x and y, let $\gcd(x, y)$ denote the ***greatest common divisor*** of x and y. Similarly, for two polynomials $f(x)$ and $g(x)$, let $\gcd(f(x), g(x))$ denote the ***greatest common divisor*** of $f(x)$ and $g(x)$. For two positive integers x and y, let $\ell cm(x, y)$ denote the ***least common multiple*** of x and y. When two sequences $\boldsymbol{a}$ and $\boldsymbol{b}$, where $\ell(\boldsymbol{a}) = \pi(\boldsymbol{a})$ and $\ell(\boldsymbol{b}) = \pi(\boldsymbol{b})$ are not of the same period, the bitwise product $\boldsymbol{a} \cdot \boldsymbol{b}$ is defined as the bitwise product of the sequences $(\overbrace{\boldsymbol{a}\boldsymbol{a} \cdots \boldsymbol{a}}^{\mu_1 \text{ copies of } \boldsymbol{a}})$ and $(\overbrace{\boldsymbol{b}\boldsymbol{b} \cdots \boldsymbol{b}}^{\mu_2 \text{ copies of } \boldsymbol{b}})$, where $\mu_1 = \frac{\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))}{\pi(\boldsymbol{a})}$ and $\mu_2 = \frac{\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))}{\pi(\boldsymbol{b})}$. This implies that the sequence $\boldsymbol{a} \cdot \boldsymbol{b}$ has length and period $\ell cm(\pi(\boldsymbol{a}), \pi(\boldsymbol{b}))$. In the same way, we define the bitwise addition of $\boldsymbol{a} + \boldsymbol{b}$.

The following results appear in Chapter 2 of [8] or can be inferred from the discussion in this chapter. The first result is a simple observation, but its consequences are of considerable importance throughout the paper.

Proposition 1: If $\boldsymbol{a}$ and $\boldsymbol{b}$ are two sequences generated by the polynomial $f(x)$, then the sequence $\boldsymbol{a}+\boldsymbol{b}$ is also generated by the polynomial $f(x)$.

Proof: The claim follows immediately from the fact that if $\boldsymbol{a}$ and $\boldsymbol{b}$ satisfy the same linear recurrence, then $\boldsymbol{a} + \boldsymbol{b}$ also satisfies the same recurrence. ∎

The property implied by Proposition 1 is called ***the shift-and-add property*** for sequences generated by the same LFSR, i.e., the same polynomial.

Corollary 1: The sequences generated by a polynomial $f(x)$ are closed under bitwise addition.

C. The Exponent of a Polynomial

Definition 6: The ***exponent*** $e(h(x))$ of a polynomial $h(x)$, where $h(0) \neq 0$, is the least positive integer e such that $h(x)$ divides $x^e - 1$. (Some authors call this the ***order*** of the polynomial.)

Let $\gamma(x)$ be the polynomial defined by

$$\gamma(x) = \sum_{i=1}^{n} c_i x^i (a_{-i} x^{-i} + \cdots + a_{-1} x^{-1}) = \sum_{i=0}^{n-1} \gamma_i x^i.$$

Proposition 2: If the sequence (a_k) is a nonzero sequence generated by the characteristic polynomial $c(x)$ such that $\gcd(\gamma(x), c(x)) = 1$, then $\pi((a_k)) = e(c(x))$.

Corollary 2: If $c(x)$ is an irreducible polynomial, then the least period of its associated sequence (a_k) is the same for each initial state, except for the all-zero state.

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Corollary 3: If $c(x)$ is an irreducible polynomial of degree n, then all nonzero sequences it generates have the same period, which is a factor of $2^n - 1$.

Definition 7: An irreducible polynomial of degree n which is a characteristic polynomial of an LFSR_n that generates a sequence of period $2^n - 1$, is called a ***primitive polynomial***. The sequence of period $2^n - 1$ that it generates is called a span n **M-*sequence***.

Definition 8: A ***zero factor*** $\mathrm{ZF}(n, k)$ with ***exponent*** k is a set of d cyclic sequences of least period k, which contains each nonzero n-tuple exactly once as a window in one of the cycles.

For a zero factor with d cycles of length k, we must have $d \cdot k = 2^n - 1$ and $n < k \leq 2^n - 1$. All known parameters, n, d, and k, can be inferred from the following result proved in [8] and [14].

Proposition 3: Let $f_i(x)$, $1 \leq i \leq t$, be t distinct irreducible polynomials of degree n and exponent e. Then the feedback shift register which has the characteristic polynomial $\prod_{i=1}^{t} f_i(x)$ produces a zero factor with exponent e.

Definition 9: A reducible polynomial has a ***uniform exponent*** e if it is a product of distinct irreducible polynomials of exponent e, i.e., all nonzero sequences it generates have the same period e.

All zero factors whose sequences are generated by polynomials are induced by Proposition 3. The following proposition and its corollary were proved in [20, p. 85, Theorem 3.5].

Proposition 4: The degree d of an irreducible polynomial with exponent e is equal to the multiplicative order of 2 modulo e, i.e., d is the smallest integer such that $2^d \equiv 1 \pmod{e}$.

Corollary 4: All the irreducible polynomials with exponent e have the same degree n. All reducible polynomials with a uniform exponent e are products of irreducible polynomials of the same degree n and exponent e.

All the reducible polynomials that are discussed in the paper have a uniform exponent, so when we say 'a polynomial of exponent e' we always mean 'a polynomial of uniform exponent e'.

Definition 10: For a polynomial $f(x)$, let $S(f(x))$ denote the set of cyclic sequences whose characteristic polynomial is $f(x)$. The polynomial $f(x)$ ***generates*** these sequences and each sequence $\boldsymbol{a}$ in $S(f(x))$ is ***generated*** by $f(x)$. Note that the all-zero sequence is always generated by any given polynomial, but sometimes it will be excluded. This will be explicitly mentioned or clearly understood from the context.

Corollary 5: Let $f_i(x)$, $1 \leq i \leq t$, be t distinct irreducible polynomials of degree n and exponent e. The polynomial $\prod_{i=1}^{t} f_i(x)$ generates all the sequences generated by the polynomial $\prod_{i=1}^{t-1} f_i(x)$ Each $((t-1)n)$-tuple is contained as a window in exactly one sequence generated by $\prod_{i=1}^{t-1} f_i(x)$. Each (tn)-tuple is contained as a window in exactly one sequence generated by $\prod_{i=1}^{t} f_i(x)$.

D. Irreducible Polynomials

Sometimes it will be important to distinguish between primitive polynomials and irreducible polynomials that are not primitive, and hence such an irreducible polynomial will be called an ***irreducible non-primitive polynomial*** (in short, ***INP polynomial***).

Let $c(x)$ be an irreducible polynomial. Consider the irreducible polynomial

$$\hat{c}(x) \triangleq x^n c(x^{-1}) = x^n - \sum_{i=1}^{n} c_i x^{n-i}, \tag{2}$$

called the ***companion polynomial***, and the ***companion matrix*** C of $c(x)$ defined by

$$C = \begin{bmatrix} 0 & 0 \cdots 0 & c_n \\ 1 & 0 \cdots 0 & c_{n-1} \\ 0 & 1 \cdots 0 & c_{n-2} \\ \vdots & \vdots \ddots \vdots & \vdots \\ 0 & 0 \cdots 1 & c_1 \end{bmatrix}.$$

If β is a root of $\hat{c}(x)$ whose order is t, then it can be proved in a similar way to [8, pp. 66–67] that we can order the powers of β, using a column-vector representation of a finite field with β^i for $0 \leq i < n$ as a basis, by

$$(\beta^0), (\beta^1), (\beta^2), \ldots, (\beta^{t-1}) .$$

III. Folding a Sequence Into an Array

In this section, we will describe the folding technique defined by MacWilliams and Sloane [22]. The technique was used in [22] to fold an M-sequence into an array, and if the array satisfies certain conditions, then the result is a PRA. We will generalize the proof and show that these conditions can be weakened. In Section VII, we prove that the same conditions suffice to show that a set of arrays forms a PRAC.

A. Folding and a Basic Theorem

The ***folding*** technique which is going to be used to construct an $(r_1, r_2; n_1, n_2)$-PRA in this section (and also an $(r_1, r_2; n_1, n_2)$-PRAC in the other sections) is based on folding nonzero sequences generated by a polynomial that generates a zero factor with exponent $r_1 r_2$ into $r_1 \times r_2$ arrays. Assume that $\eta = 2^{n_1 n_2} - 1$, $r_1 = 2^{n_1} - 1$, and $r_2 = \frac{\eta}{r_1}$, where $\gcd(r_1, r_2) = 1$. Let $\mathcal{S} = [s_0 s_1 s_2 \cdots s_{\eta-1}]$ be a span $n_1 n_2$ M-sequence. Write the consecutive elements of $\mathcal{S}$ down the southeast diagonals of an $r_1 \times r_2$ array $\mathcal{B} = \{b_{ij}\}$, $0 \leq i \leq r_1 - 1$, $0 \leq j \leq r_2 - 1$, starting at b_{00}, b_{11}, b_{22} and so on, where the last position is b_{r_1-1, r_2-1}. After writing position b_{ij} we continue to write $b_{i+1, j+1}$, where $i+1$ is taken modulo r_1 and $j+1$ is taken modulo r_2. In other words, s_k will be written in position b_{ij}, where $i \equiv k \pmod{r_1}$ and $j \equiv k \pmod{r_2}$. By the Chinese Remainder Theorem [8, pp. 11–13] each k, where $0 \leq k \leq r_1 r_2 - 1$, has a different solution for these two equations and so the diagonal covers the entire array. Another type of folding to generate pseudo-random arrays was presented in [36]. A generalization of the technique by folding sequences into various shapes was considered and analyzed in [7]. The following theorem was proved in [22].

Theorem 1: Each $n_1 \times n_2$ nonzero matrix is contained exactly once as a window in the $r_1 \times r_2$ array $\mathcal{B}$ obtained by folding a span $n_1 n_2$ M-sequence, where $r_1 = 2^{n_1} - 1$ and $r_2 = (2^{n_1 n_2} - 1)/r_1$. Moreover, $\mathcal{B}$ has the shift-and-add property, that is, $\mathcal{B}$ is an $(r_1, r_2; n_1, n_2)$-PRA.

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The proof of Theorem 1 presented in [22] is based on the observation that it is sufficient to prove that in the upper-left $n_1 \times n_2$ window of the array $\mathcal{B}$ we cannot have the all-zero $n_1 \times n_2$ matrix.

If the length of $\mathcal{S}$ is $r_1 r_2$, where $\gcd(r_1, r_2) = 1$ we denote by $\mathcal{F}(\mathcal{S}; r_1, r_2)$ the matrix $\mathcal{B}$ of size $r_1 \times r_2$ obtained by folding $\mathcal{S}$ into $\mathcal{B}$.

Example 1: For $n_1 = n_2 = 2$, $r_1 = 3$, $r_2 = 5$, consider the span 4 M-sequence $\mathcal{S} = [000111101011001]$, with positions numbered from 0, 1, up to 14. Consider now the 3×5 array $\mathcal{B}$ with the entries b_{ij}, $0 \le i \le 2$, $0 \le j \le 4$, where the positions 0 through 14, of the sequence, are folded into $\mathcal{B}$ as follows

$$\begin{bmatrix} b_{00} & b_{01} & b_{02} & b_{03} & b_{04} \\ b_{10} & b_{11} & b_{12} & b_{13} & b_{14} \\ b_{20} & b_{21} & b_{22} & b_{23} & b_{24} \end{bmatrix}, \quad \begin{bmatrix} 0 & 6 & 12 & 3 & 9 \\ 10 & 1 & 7 & 13 & 4 \\ 5 & 11 & 2 & 8 & 14 \end{bmatrix}.$$

The M-sequence $\mathcal{S}$ is folded into the array $\mathcal{B}$ to produce the array,

$$\mathcal{F}(\mathcal{S}; 3, 5) = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

which forms a $(3, 5; 2, 2)$-PRA. ∎

If we shift the array $\mathcal{B}$ horizontally and/or vertically, we obtain an array that is a folding of a shift of $\mathcal{S}$. Since the sequences obtained from any primitive polynomial have the shift-and-add property, it follows that if we add $\mathcal{B}$ to one of its non-constant shifts, the outcome is another non-constant shift of the array $\mathcal{B}$. This property is the shift-and-add property of the array. This property is also derived by observing that the folding operation preserves the operation of bitwise addition of sequences. A similar property is satisfied for the bitwise product. This is summarized in the following lemma.

Lemma 2: If $\boldsymbol{u}$ and $\boldsymbol{v}$ are sequences of length $r_1 r_2$ such that $\gcd(r_1, r_2) = 1$ then

1) $\mathcal{F}(\boldsymbol{u}; r_1, r_2) + \mathcal{F}(\boldsymbol{v}; r_1, r_2) = \mathcal{F}(\boldsymbol{u} + \boldsymbol{v}; r_1, r_2)$;
2) $\mathcal{F}(\boldsymbol{u}; r_1, r_2) \cdot \mathcal{F}(\boldsymbol{v}; r_1, r_2) = \mathcal{F}(\boldsymbol{u} \cdot \boldsymbol{v}; r_1, r_2)$.

Example 2: Consider the array and the M-sequence $\mathcal{S}$ of Example 1. We shift the array horizontally by 2 and vertically by 1 and add them as follows, where the first bit of $\mathcal{S}$ is in bold.

$$\begin{bmatrix} \mathbf{0}1010 \\ 10001 \\ 11011 \end{bmatrix} + \begin{bmatrix} 11110 \\ 10\mathbf{0}10 \\ 01100 \end{bmatrix} = \begin{bmatrix} 1010\mathbf{0} \\ 00011 \\ 10111 \end{bmatrix}.$$

The M-sequence $\mathcal{S}$ starts in the leftmost array in b_{00}, in the middle array at b_{12}, and in the rightmost array at b_{04}. ∎

Definition 11: A ***constant polynomial*** is a polynomial whose degree is zero (and the zero polynomial is also a constant polynomial). The ***minimal polynomial*** $m(x)$ of an element α in a finite field $\mathbb{F}_{2^n}$ is the monic polynomial over $\mathbb{F}_{2^t}$ of the least degree such that $m(\alpha) = 0$, where $\mathbb{F}_{2^t}$ is a subfield of $\mathbb{F}_{2^n}$.

B. A Generalised Theorem

Theorem 1 is generalized to obtain more PRAs, with our first main technique, as follows.

Theorem 2: Let n_1, n_2, r_1, and r_2, be positive integers such that $r_1 r_2 = 2^{n_1 n_2} - 1$ and $\gcd(r_1, r_2) = 1$. If r_1 divides $2^{n_1} - 1$ and the n_1 integers, 2^i, $0 \le i \le n_1 - 1$, are distinct modulo r_1, then folding an M-sequence whose length is $r_1 r_2$ into an $r_1 \times r_2$ array $\mathcal{B}$ forms an $(r_1, r_2; n_1, n_2)$-PRA.

Proof: Let α be a primitive element in $\mathbb{F}_{2^{n_1 n_2}}$ and let $\mathcal{S}$ be its associated M-sequence. Furthermore, let G be an $(n_1 n_2) \times (2^{n_1 n_2} - 1)$ matrix defined by the $n_1 n_2$ consecutive shifts of the M-sequence $\mathcal{S}$. In such a representation, the columns of G are arranged in the order isomorphic to $\alpha^0, \alpha^1, \alpha^2, \ldots, \alpha^{2^{n_1 n_2} - 2}$ defined by the companion matrix. Let $\ell(i, j)$ be the integer with $0 \le \ell(i, j) \le 2^{n_1 n_2} - 1$ such that $\ell(i, j) \equiv i \pmod{r_1}$ and $\ell(i, j) \equiv j \pmod{r_2}$.

We claim that the $n_1 n_2$ columns of G associated with the elements $\alpha^{\ell(i,j)}$, where $0 \le i \le n_1 - 1$ and $0 \le j \le n_2 - 1$, are linearly independent. These columns are associated with the $n_1 \times n_2$ sub-array in the upper-left corner of $\mathcal{B}$.

Assume for a contradiction that

$$\sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} c_{ij} \alpha^{\ell(i,j)} = 0,$$

where not all the c_{ij} are *zero*. Since $\gcd(r_1, r_2) = 1$, it follows by Bézout's identity [8, pp. 7–9] that there exist two integers μ and ν such that $\mu r_1 + \nu r_2 = 1$. Let $\beta = \alpha^{\nu r_2}$ and $\gamma = \alpha^{\mu r_1}$. We show that β is an element of order r_1. To see this, assume for a contradiction that $\beta^t = 1$ for $t < r_1$, i.e., $\beta^t = \alpha^{\nu r_2 t} = 1$. The order of α is $r_1 r_2$ and hence $r_1 r_2$ divides $\nu r_2 t$, i.e., r_1 divides νt. Since $\mu r_1 + \nu r_2 = 1$ we have that $\gcd(r_1, \nu) = 1$ which implies that r_1 divides t, i.e., $r_1 \le t$, a contradiction. Hence, β has order r_1 as required. Similarly, γ is an element of order r_2. Now, $\alpha = \alpha^{\mu r_1 + \nu r_2} = \beta\gamma$ and hence $\alpha^{\ell(i,j)} = \beta^{\ell(i,j)} \gamma^{\ell(i,j)} = \beta^i \gamma^j$. Therefore, we have

$$\begin{aligned} 0 &= \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} c_{ij} \alpha^{\ell(i,j)} \\ &= \sum_{i=0}^{n_1-1} \sum_{j=0}^{n_2-1} c_{ij} (\beta\gamma)^{\ell(i,j)} \\ &= \sum_{j=0}^{n_2-1} \left(\sum_{i=0}^{n_1-1} c_{ij} \beta^i \right) \gamma^j. \end{aligned} \tag{3}$$

We claim that $\sum_{i=0}^{n_1-1} c_{i,j} \beta^i = 0$ for all j.

Since $\beta^{r_1} = 1$ and r_1 divides $2^{n_1} - 1$, we have that $\beta \in \mathbb{F}_{2^{n_1}}$ and hence $\sum_{i=0}^{n_1-1} (c_{i,j} \beta^i) \in \mathbb{F}_{2^{n_1}}$, i.e., the coefficient of γ^j in Equation (3) is an element of $\mathbb{F}_{2^{n_1}}$.

Let m be the smallest positive integer such that $\gamma^{2^m - 1} = 1$. Clearly, $\gamma^{2^m - 1} = \alpha^{\mu r_1 (2^m - 1)} = 1$ and since the order of α is $r_1 r_2$, it follows that $r_1 r_2$ divides $\mu r_1 (2^m - 1)$ and hence r_2 divides $\mu(2^m - 1)$. Moreover, since $\mu r_1 + \nu r_2 = 1$, it follows that $\gcd(\mu, r_2) = 1$ and therefore r_2 divides $2^m - 1$. We have that $r_2 = \frac{2^{n_1 n_2} - 1}{r_1} = \frac{2^{n_1 n_2} - 1}{2^{n_1} - 1} d$, where $d = \frac{2^{n_1} - 1}{r_1}$. Since r_2 divides $2^m - 1$, it follows that $\frac{2^{n_1 n_2} - 1}{2^{n_1} - 1} d$ divides $2^m - 1$ and hence $\frac{2^{n_1 n_2} - 1}{2^{n_1} - 1}$ divides $2^m - 1$ and d divides $2^m - 1$.

The binary representation of $\frac{2^{n_1 n_2} - 1}{2^{n_1} - 1}$ (which divides $2^m - 1$) is $10^{n_1 - 1} 10^{n_1 - 1} 1 \cdots 0^{n_1 - 1} 1$, where the number of *ones* in this representation is n_2. Hence, this binary representation contains $n_1(n_2 - 1) + 1$ digits. The binary representation of $2^m - 1$ is $11 \cdots 1$, where the number of *ones* in this representation is m. Hence, by considering binary multiplication, the smallest m

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for which r_2 divides $2^m - 1$ is $n_1 n_2$. Thus, $m = n_1 n_2$ is the smallest positive integer such that $\gamma^{2^m} = \gamma$, the n_2 elements

$$\gamma, \gamma^{2^{n_1}}, \gamma^{2^{2n_1}}, \ldots, \gamma^{2^{(n_2-1)n_1}}$$

are distinct. Therefore,

$$\prod_{i=0}^{n_2-1} (x - \gamma^{2^{in_1}})$$

is the minimal polynomial of γ in $\mathbb{F}_{2^{n_1}}$. This polynomial has degree n_2 in $\mathbb{F}_{2^{n_1}}$. Now, the polynomial on the right side of Equation (3) is a polynomial in γ with a smaller degree $n_2 - 1$ which is equal to *zero*.

This implies that all the coefficients of γ^j in Equation (3) are equal to *zero*. Therefore, for each $0 \le j \le n_2 - 1$ we have

$$\sum_{i=0}^{n_1-1} c_{ij} \beta^i = 0,$$

as claimed. But this implies that the coefficients c_{ij} are all zero, since this is a polynomial in β of degree less than n_1 and the minimal polynomial of β over $\mathbb{F}_2$ has degree n_1 (since r_1 divides $2^{n_1} - 1$ we have that $\beta^{2^{n_1}} = \beta$. Moreover, β^{2^0}, β^{2^1},...,$\beta^{2^{n_1-1}}$, are distinct and hence β is a zero of an irreducible polynomial of degree n_1 over $\mathbb{F}_2$ and this is its minimal polynomial.) But this is a contradiction, since we are assuming that not all the elements c_{ij} are zero. This completes the proof of the claim that the $n_1 n_2$ columns of G associated with the elements $\alpha^{\ell(i,j)}$, where $\ell(i, j) \equiv i \pmod{r_1}$, $\ell(i, j) \equiv j \pmod{r_2}$, $0 \le i \le n_1 - 1$ and $0 \le j \le n_2 - 1$, are linearly independent.

This claim implies that the $n_1 \times n_2$ array in the upper-left corner of $\mathcal{B}$ is nonzero. This $n_1 \times n_2$ window can be chosen arbitrarily since the M-sequence can start at any nonzero initial $(n_1 n_2)$-tuple. This $n_1 \times n_2$ window determines the rest of the elements in $\mathcal{B}$. Hence, by the shift-and-add property, there are no two equal $n_1 \times n_2$ windows, as otherwise we can have an all-zero $n_1 \times n_2$ window by adding the associated two such shifts with two equal $n_1 \times n_2$ windows. Thus, we have the window property. ■

C. Small Parameter Sets

For $r_1 r_2 \le 511$, there are 9 possible parameters for $(r_1, r_2; n_1, n_2)$-PRAs obtained via folding which implies that $\gcd(r_1, r_2) = 1$. A $(3, 5; 2, 2)$-PRA can be constructed via both Theorem 1 and Theorem 2. $(7, 9; 3, 2)$-PRA can be constructed via both Theorems 1 and 2. A $(7, 9; 2, 3)$-PRA cannot be obtained via folding. A $(15, 17; 4, 2)$-PRA and a $(3, 85; 2, 4)$-PRA can be constructed via both Theorems 1 and 2. A $(5, 51; 4, 2)$-PRA can be constructed via Theorem 2. A $(5, 51; 2, 4)$-PRA and a $(15, 17; 2, 4)$-PRA exist but cannot be obtained by Theorem 2. Finally, a $(7, 73; 3, 3)$-PRA can be constructed via both Theorems 1 and 2.

IV. Folding Sequences of a Zero Factor

A. Folding a Set of Sequences

Folding one M-sequence is generalized to folding all the (nonzero) cyclic sequences of a zero factor. An introduction and analysis of such a folding is presented in this section. Let $\mathcal{C}$ be a set of sequences of length $r_1 r_2$, such that $\gcd(r_1, r_2) = 1$. We denote by $\mathcal{F}(\mathcal{C}; r_1, r_2)$ the set of $r_1 \times r_2$ arrays obtained by folding the sequences in $\mathcal{C}$ into $r_1 \times r_2$ arrays.

Example 3: Let $r_1 = 3$, $r_2 = 7$, $n_1 = 2$, and $n_2 = 3$. Let $f(x) = x^6 + x^5 + x^4 + x^2 + 1$ be the irreducible polynomial of degree 6 and exponent 21. It generates a ZF(6, 21) $\mathcal{C}$. The nonzero sequences of $\mathcal{C}$ generated by $f(x)$ are

$$[000001010010011001011],$$
$$[010000111101101010111],$$
$$[001000110111111001110].$$

Folding these three sequences of $\mathcal{C}$ into 3×7 arrays yields the following three arrays

$$\mathcal{F}(\mathcal{C}; 3, 7) = \left\{ \begin{bmatrix} 0000000 \\ 1001011 \\ 1001011 \end{bmatrix}, \begin{bmatrix} 0010111 \\ 1110010 \\ 1100101 \end{bmatrix}, \begin{bmatrix} 0010111 \\ 1001011 \\ 1011100 \end{bmatrix} \right\}.$$

It is easy to verify that these three arrays form a $(3, 7; 2, 3)$-PRAC. ■

B. Properties of the Constructed Arrays

It is readily verified that each row of the arrays in Example 3 is either the M-sequence [0010111] or the all-zero sequence. Each column is either the M-sequence [011] or the all-zero sequence. This phenomena will be explained in Section VI. Meanwhile, the following lemma, which is used to prove this property, is easily verified.

Lemma 3: Let r_1 and r_2 be two positive integers such that $\gcd(r_1, r_2) = 1$ and let $\mathcal{S} = [\boldsymbol{a}\, \boldsymbol{a} \ldots \boldsymbol{a}]$ be a sequence of length $r_1 r_2$.

(1) If the length of $\boldsymbol{a}$ is r_2, then each row of $\mathcal{F}(\mathcal{S}; r_1, r_2)$ is given by $\boldsymbol{a}$ and each column is either the all-one sequence or the all-zero sequence.
(2) If the length of $\boldsymbol{a}$ is r_1, then each column of $\mathcal{F}(\mathcal{S}; r_1, r_2)$ is given by $\boldsymbol{a}$ and each row is either the all-one sequence or the all-zero sequence.

Proof: Part (1) follows from the fact that the length of a row in $\mathcal{F}(\mathcal{S}; r_1, r_2)$ is the length of the sequence $\boldsymbol{a}$ and the consecutive bits of the sequence are written column by column starting from the first column. Therefore, the i-th bit of the sequence $\boldsymbol{a}$ is always written to the i-th column of $\mathcal{F}(\mathcal{S}; r_1, r_2)$.

Part (2) follows from the fact that the length of a column in $F(\mathcal{S}; r_1, r_2)$ is the length of the sequence $\boldsymbol{a}$ and the consecutive bits of the sequence are written row by row, starting from the first row. Therefore, the i-th bit of the sequence $\boldsymbol{a}$ is always written to the i-th row of $\mathcal{F}(\mathcal{S}; r_1, r_2)$. ■

Example 4: If $\mathcal{S} = [011\,011\,011\,011\,011\,011\,011] \in \mathbb{F}_2^{3\cdot 7}$ and $\boldsymbol{a} = [011]$, then

$$\mathcal{A} \triangleq \mathcal{F}(\mathcal{S}; 3, 7) = \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 1\,1\,1\,1\,1\,1\,1 \\ 1\,1\,1\,1\,1\,1\,1 \end{bmatrix}.$$

Example 5: If $\mathcal{S} = [1001011\,1001011\,1001011] \in \mathbb{F}_2^{3\cdot 7}$ and $\boldsymbol{a} = [1001011]$, then

$$\mathcal{B} \triangleq \mathcal{F}(\mathcal{S}; 3, 7) = \begin{bmatrix} 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \\ 1\,0\,0\,1\,0\,1\,1 \end{bmatrix}.$$

■

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C. Bitwise Product and Addition

Lemma 4: Let $\mathcal{A}$ be an $r_1 \times r_2$ array, where each row is given by a sequence $\boldsymbol{u}$ and each column is either the all-one sequence or the all-zero sequence. Let $\mathcal{B}$ be an $r_1 \times r_2$ array, where each column is given by a sequence $\boldsymbol{v}$ and each row is either the all-one sequence or the all-zero sequence. Then the bitwise product $\mathcal{A} \cdot \mathcal{B}$ is an $r_1 \times r_2$ array, where each row is either an all-zero sequence or the sequence $\boldsymbol{u}$, and each column of $\mathcal{A} \cdot \mathcal{B}$ is either an all-zero sequence or the sequence $\boldsymbol{v}$.

Proof: If the i-th row of $\mathcal{B}$ is an all-zero sequence, then the bitwise product yields an all-zero sequence and hence the i-th row of $\mathcal{A} \cdot \mathcal{B}$ is the all-zero sequence. If the i-th row of $\mathcal{B}$ is an all-one sequence, then the bitwise product with the sequence $\boldsymbol{u}$ yields the sequence $\boldsymbol{u}$ and hence the i-th row of $\mathcal{A} \cdot \mathcal{B}$ is the sequence $\boldsymbol{u}$.

If the i-th column of $\mathcal{A}$ is an all-zero sequence, then the bitwise product yields an all-zero sequence and hence the i-th column of $\mathcal{A} \cdot \mathcal{B}$ is the all-zero sequence. If the i-th column of $\mathcal{B}$ is an all-one sequence, then the bitwise product with the sequence $\boldsymbol{v}$ yields the sequence $\boldsymbol{v}$ and hence the i-th row of $\mathcal{A} \cdot \mathcal{B}$ is the sequence $\boldsymbol{v}$. ∎

Example 6: Let $\mathcal{A}$ and $\mathcal{B}$ be the 3×7 arrays of Example 4 and 5, respectively. The bitwise product of $\mathcal{A}$ and $\mathcal{B}$, $\mathcal{A} \cdot \mathcal{B}$, is as follows

$$\mathcal{A} \cdot \mathcal{B} = \begin{bmatrix} 0000000 \\ 1111111 \\ 1111111 \end{bmatrix} \cdot \begin{bmatrix} 1001011 \\ 1001011 \\ 1001011 \end{bmatrix} = \begin{bmatrix} 0000000 \\ 1001011 \\ 1001011 \end{bmatrix}.$$

∎

For the rest of this section let $f_1(x)$ be a polynomial which forms a zero factor with exponent r_1 and $f_2(x)$ be a polynomial which forms a zero factor with exponent r_2. Recall Definition 10, the set $S(f(X)$ of all sequences with characteristic polynomial $f(x)$.

Lemma 5: If $\mathcal{A}$ and $\mathcal{B}$ are two distinct $r_1 \times r_2$ arrays whose columns are either all-zero sequences or a sequences of length r_1 contained in $S(f_1(x))$ and whose rows are either all-zero sequences or sequences of length r_2 contained in $S(f_2(x))$, then the addition $\mathcal{A}+\mathcal{B}$ is also an $r_1 \times r_2$ array whose columns are either all-zero sequences or sequences contained in $S(f_1(x))$, and whose rows are either all-zero sequences or sequences contained in $S(f_2(x))$.

Proof: Since the columns of $\mathcal{A}$ and $\mathcal{B}$ (including those which are all-zero sequences) are sequences generated by $f_1(x)$, it follows by Proposition 1 that their addition is also a sequence generated by $f_1(x)$. Hence, each column of $\mathcal{A}+\mathcal{B}$ is either an all-zero sequence or a sequence contained in $S(f_1(x))$. Similarly, a row of $\mathcal{A}+\mathcal{B}$ is either an all-zero sequence or a sequence contained in $S(f_2(x))$. ∎

Example 7: Consider the primitive polynomial $f_1(x) = x^2 + x + 1$ whose zero factor has exactly one sequence [011] and the reducible polynomial $f_2(x) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = (x^3 + x^2 + 1)(x^3 + x + 1)$ whose zero factor $\mathcal{C}$ has nine sequences [0011101], [0010111], [0000101], [0101011], [0001111], [0111111], [0001001], [0000011], [0011011]. In the following 3×7 array $\mathcal{A}$, each column is either the all-zero sequence or the sequence [011] (or a shift of it). Each row is either the all-zero sequence or the sequence [0001111] (or a shift of it).

$$\mathcal{A} \triangleq \begin{bmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 0\,0\,0\,1\,1\,1\,1 \\ 0\,0\,0\,1\,1\,1\,1 \end{bmatrix}.$$

Adding $\mathcal{A}$ to one of its shifts $\mathcal{B}$ we obtain the following array $\mathcal{T}_1$:

$$\mathcal{T}_1 \triangleq \mathcal{A} + \mathcal{B} = \begin{bmatrix} 0000000 \\ 0001111 \\ 0001111 \end{bmatrix} + \begin{bmatrix} 0011110 \\ 0011110 \\ 0000000 \end{bmatrix} = \begin{bmatrix} 0011110 \\ 0010001 \\ 0001111 \end{bmatrix}.$$

Adding $\mathcal{T}_1$ to one of its shifts $\mathcal{T}_2$ we obtain the following array:

$$\mathcal{T}_1 + \mathcal{T}_2 = \begin{bmatrix} 0011110 \\ 0010001 \\ 0001111 \end{bmatrix} + \begin{bmatrix} 0111100 \\ 1111000 \\ 1000100 \end{bmatrix} = \begin{bmatrix} 0100010 \\ 1101001 \\ 1001011 \end{bmatrix}.$$

Each column of $\mathcal{A}+\mathcal{B}$ and of $\mathcal{T}_1+\mathcal{T}_2$ is either the all-zero sequence or the sequence [011], and each row is either the all-zero sequence or a sequence from $\mathcal{C}$. ∎

Let $S(f_1(x))$ and $S(f_2(x))$ be the set of nonzero sequences generated by $f_1(x)$ and $f_2(x)$, respectively (these sequences are taken in all their shifts). Assume that all the sequences in $S(f_i(x))$ have least period r_i, $i = 1, 2$ and $\gcd(r_1, r_2) = 1$. Let

$$\mathbb{C} \triangleq \{\mathcal{F}(\boldsymbol{a} \cdot \boldsymbol{b}; r_1, r_2) \ : \ \boldsymbol{a} \in S(f_1(x)), \quad \boldsymbol{b} \in S(f_2(x))\}.$$

Lemma 6: Let $\boldsymbol{a}$ be a nonzero sequence generated by $f_1(x)$ and let $\boldsymbol{b}$ be a nonzero sequence generated by $f_2(x)$. If $\mathcal{S}_1 \triangleq [\boldsymbol{a}\,\boldsymbol{a} \cdots \boldsymbol{a}] \in \mathbb{F}_2^{r_1 \cdot r_2}$ and $\mathcal{S}_2 \triangleq [\boldsymbol{b}\,\boldsymbol{b} \cdots \boldsymbol{b}] \in \mathbb{F}_2^{r_1 \cdot r_2}$, then the $r_1 \times r_2$ array $\mathcal{F}(\mathcal{S}_1; r_1, r_2) \cdot \mathcal{F}(\mathcal{S}_2; r_1, r_2)$ is contained in $\mathbb{C}$.

Proof: The claim in the lemma follows directly from Lemma 2. ∎

Corollary 6: If $\tilde{\boldsymbol{a}}$ is an n_1-tuple in the sequence $\boldsymbol{a}$ and $\tilde{\boldsymbol{b}}$ is an n_2-tuple in the sequence $\boldsymbol{b}$, then the $n_1 \times n_2$ matrix M whose i-th row is $\tilde{\boldsymbol{b}}$ if the i-th entry of $\tilde{\boldsymbol{a}}$ is *one* and all-zero if the i-th entry of $\tilde{\boldsymbol{a}}$ is *zero*, is contained in $\mathcal{F}(\boldsymbol{a} \cdot \boldsymbol{b}; r_1, r_2)$.

Corollary 7: If $\tilde{\boldsymbol{a}}$ is an n_1-tuple in the sequence $\boldsymbol{a}$ and $\tilde{\boldsymbol{b}}$ is an n_2-tuple in the sequence $\boldsymbol{b}$, then the $n_1 \times n_2$ matrix M whose i-th column is $\tilde{\boldsymbol{a}}$ if the i-th entry of $\tilde{\boldsymbol{b}}$ is *one* and all-zero if the i-th entry of $\tilde{\boldsymbol{b}}$ is *zero*, is contained in $\mathcal{F}(\boldsymbol{a} \cdot \boldsymbol{b}; r_1, r_2)$.

Definition 12: Let $S(f_1(x)) \cdot S(f_2(x))$ be defined as a vector space spanned by all products $\boldsymbol{a} \cdot \boldsymbol{b}$, where $\boldsymbol{a} \in S(f_1(x))$ and $\boldsymbol{b} \in S(f_2(x))$.

For the following lemma, we are not aware of a proof in [20], although we assume that the authors were aware of the result. The lemma can be proved by careful enumeration (see the proof in [8, pp. 61–63, Lemma 2.5 and Theorem 2.5]). It is also a corollary of Proposition 5 below.

Lemma 7: Let $S(f_1(x))$ and $S(f_1(x))$ be the set of nonzero sequences generated by $f_1(x)$ and $f_2(x)$, respectively. Assume that all the sequences in $S(f_i(x))$ have least period r_i, $i = 1, 2$. If $\gcd(r_1, r_2) = 1$, then the least period of the nonzero sequences in the linear span of the sequences in the set

$$\mathbb{C} \triangleq \{\boldsymbol{a} \cdot \boldsymbol{b} \ : \ \boldsymbol{a} \in S(f_1(x)), \quad \boldsymbol{b} \in S(f_2(x))\}$$

is $r_1 r_2$.

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V. A Necessary and Sufficient Condition for Irreducible Polynomials

The proof in [22] that the array $\mathcal{B}$ is an $(r_1, r_2; n_1, n_2)$-PRA requires that $r_1 = 2^{n_1} - 1$. It appears that this requirement is necessary in some cases, but it is not required in other cases (see Theorem 2). The proof of Theorem 1 given in [22] and also the proof of Theorem 2 is based on the observation that it is sufficient to show that the $n_1 \times n_2$ all-zero matrix is not contained as a window of the array $\mathcal{B}$. In this section, a necessary and sufficient condition that an array (or set of arrays) constructed by folding sequences generated by an irreducible polynomial is a pseudo-random array (or a pseudo-random array code) will be given. This condition is key for constructing such arrays and array codes, respectively, but can be inefficient to verify computationally. Later in the paper, more direct (but less general) conditions will be presented for the families of codes constructed in the paper. Section IX presents an alternative, more efficient, general condition. All these conditions can be also used for VLSI testing.

A. VLSI Testing

For this purpose, we develop a simple theory. The concepts of this theory are similar to the concepts developed for VLSI testing in Lempel and Cohn [19]. However, the main proof provided here is different since the proof in [19] does not hold in the more general situation in which we will be working. In VLSI testing, a test sequence $\mathbf{s}$ is called (s,t)-***universal*** if it exercises every function depending on t or fewer inputs on a VLSI chip with s inputs. The problem has attracted a lot of interest, and various solutions have been suggested. Some of these solutions are based on FSRs, e.g., [1], [17], [19], [40], and [41].

Definition 13: For a set $P = \{p_0, p_1, \ldots, p_{t-1}\}$ of t positions in a sequence, the ***set polynomial*** $g_P(x)$ is defined by

$$g_P(x) \triangleq \prod_{Q \subseteq P} \sum_{p_i \in Q} x^{p_i} \ .$$

Let $f(x) = 1 + \sum_{i=1}^{n} c_i x^i$, where $c_n = 1$ be an irreducible polynomial with exponent e. Let $\mathcal{A}$ be the set of nonzero cyclic sequences, with characteristic polynomial $f(x)$, that satisfy the recurrence in Equation (1). So $\mathcal{A}$ consists of a single M-sequence if $f(x)$ is primitive, and consists of several sequences all of period e otherwise; see Corollary 3 and Proposition 3. Consider all the possible shifts of the nonzero sequences in $\mathcal{A}$ (generated by $f(x)$) as rows in a $(2^n - 1) \times e$ matrix B, and let T be the $(2^n - 1) \times n$ matrix which is formed by a projection of any n columns of B.

Lemma 8: Every nonzero n-tuple appears as a row of the matrix T if and only if the columns of T are linearly independent.

Proof: Assume first that each n-tuple appears as a row in T. This immediately implies that the n columns of T are linearly independent.

Assume now that the columns of T are linearly independent. Since each nonzero n-tuple appears as a window exactly once in one of the nonzero sequences generated by $f(x)$, it follows that every n consecutive columns of B contain each one of the $2^n - 1$ nonzero n-tuples as a row. Hence, the first n columns of B contain each non-zero n-tuple exactly once. These column vectors can be used as rows for the generator matrix of the simplex code of length $2^n - 1$ and dimension n. Each other column of B can be represented as a linear combination of the first n columns of B. This linear combination is defined by the recursion induced by $f(x)$ given in Equation (1). Hence, all these linear combinations coincide with the codewords of the simplex code. This implies that every n linearly independent columns contain each nonzero n-tuple as a row in T. ■

Lemma 9: Let $f(x)$ be an irreducible polynomial. If Q is a nonempty subset of P (see definition 13) and $q(x) = \sum_{p_i \in Q} x^{p_i}$, then $f(x)$ divides $q(x)$ if and only if the columns of B that are associated with the subset Q sum to zero.

Proof: If the columns in B that are associated with the subset Q sum to zero, then one of the columns is a sum of some of the other columns, that is, this column is a linear combination of the other columns. This linear combination is induced by the polynomial $f(x)$ and hence $q(\beta) = 0$, where β is a root of $f(x)$. Since we also have $f(\beta) = 0$ and $f(x)$ is an irreducible polynomial, it follows that $f(x)$ divides $q(x)$.

If $f(x)$ divides $q(x)$, then $f(\alpha) = 0$ implies that $q(\alpha) = 0$ and hence since the columns of B are defined by the recursion induced by $f(x)$, it follows that the columns of B associated with the subset Q sum to zero. ■

The given proof of Lemma 9 will not work if $f(x)$ is not an irreducible polynomial since $f(x)$ does not have to divide $q(x)$ when it shares a non-trivial factor with $q(x)$.

Theorem 3: Given an irreducible polynomial $f(x)$ and a set P of n coordinates in B, then the set P of coordinates in B contains each nonzero n-tuple if and only if $g_P(x)$ is not divisible by $f(x)$.

Proof: Consider the $(2^n - 1) \times n$ matrix T projected by the n columns of B associated with the coordinates of P. By Lemma 8 every nonzero n-tuple appears as a row in T if and only if the columns of T are linearly independent. The columns of T are linearly dependent if and only if a non-empty subset of the columns in T sums to zero.

By Lemma 9 we have that $f(x)$ divides the polynomial $\sum_{p_i \in Q} x^{p_i}$, where Q is a nonempty subset of P, if and only if the associated subset of columns of T sums to zero.

Since the polynomial $f(x)$ is irreducible, it follows that $f(x)$ divides the set polynomial $g_P(x)$ if and only if there exists a subset $Q \subseteq P$ such that $f(x)$ divides the factor $\sum_{p_i \in Q} x^{p_i}$ of $g_P(x)$. Hence, by Lemmas 8 and 9, the proof is completed. ■

Theorem 3 is a necessary and sufficient condition and hence yields a method to verify whether a sequence that is constructed by a concatenation of all nonzero sequences generated by an irreducible polynomial is (s,t)-universal. The same is true for testing all the arrays generated by folding the sequences generated by an irreducible polynomial.

B. Verifying the Arrays Obtained by Folding

To examine whether a set of arrays formed by folding is an $(r_1, r_2; n_1, n_2)$-PRAC we have to examine all the positions in the sequences associated with $n_1 \times n_2$ windows in the folding. But, since the positions in two such windows differ by a cyclic shift it follows that the set polynomials of two different positions differ by a multiple of x^i, for some i, which implies

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that exactly one set polynomial should be examined to verify whether the constructed arrays form an $(r_1, r_2; n_1, n_2)$-PRAC.

Example 8: Consider the following four INP polynomials– 1011101001111, i.e., $f_1(x) = x^{12} + x^{10} + x^9 + x^8 + x^6 + x^3 + x^2 + x + 1$; 1100101101111, i.e., $f_2(x) = x^{12} + x^{11} + x^8 + x^6 + x^5 + x^3 + x^2 + x + 1$; 1110001011111, i.e., $f_3(x) = x^{12} + x^{11} + x^{10} + x^6 + x^4 + x^3 + x^2 + x + 1$; 1010011011111, i.e., $f_4(x) = x^{12} + x^{10} + x^7 + x^6 + x^4 + x^3 + x^2 + x + 1$. These four polynomials of degree 12 are four of the 24 polynomials whose exponent is 455. Now, consider the folding of their sequences into 13×35 arrays. For each of these polynomials, we ask whether its folding contains an all-zero 4×3 array, and also whether its folding contains an all-zero 3×4 array.

We start by considering the 4×3 windows. The entries on the upper-left corner of an array are $(0, 0)$, $(0, 1)$, $(0, 2)$, $(1, 0)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 1)$, $(2, 2)$, $(3, 0)$, $(3, 1)$, and $(3, 2)$. Each entry (i, j) is translated into a position k in the sequence using the pair of equations

$$k \equiv i \pmod{13}$$
$$k \equiv j \pmod{35}$$

The 12 positions for the set polynomial $g_P(x)$ are

$$P = \{0, 1, 2, 105, 106, 143, 210, 247, 248, 351, 352, 353\}.$$

We check whether each polynomial divides $g_P(x)$, which implies that it does not produce a $(13, 35; 4, 3)$-PRAC, or it does not divide it, which implies that the folding its sequences yields a $(13, 35; 4, 3)$-PRAC.

- The polynomial $f_1(x)$ divides the factor $x^{352}+x^{315}+x^{107}+x^{106}+x^{105}+x+1$ of $g_P(x)$ and hence folding its sequences into 13×35 arrays does not yield a $(13, 35; 4, 3)$-PRAC.
- The polynomial $f_2(x)$ divides the factor $x^{352} + x^{351} + x^{211} + x^{210} + x^{106} + x^{105} + x^2 + x + 1$ of $g_P(x)$ and hence folding its sequences into 13×35 arrays does not yield a $(13, 35; 4, 3)$-PRAC.
- The polynomial $f_3(x)$ does not divide $g_P(x)$ and hence folding its sequences into 13×35 arrays yields a $(13, 35; 4, 3)$-PRAC.
- The polynomial $f_4(x)$ does not divide $g_P(x)$ and hence folding its sequences into 13×35 arrays yields a $(13, 35; 4, 3)$-PRAC.

We continue by considering the 3×4 windows and the corresponding 12 positions for the set polynomial $g_P(x)$. The positions in the upper-left corner are $(0, 0)$, $(0, 1)$, $(0, 2)$, $(0, 3)$, $(1, 0)$, $(1, 1)$, $(1, 2)$, $(1, 3)$, $(2, 0)$, $(2, 1)$, $(2, 2)$, and $(2, 3)$. Each position (i, j) is translated into a position k in the sequence using the pair of equations

$$k \equiv i \pmod{13},$$
$$k \equiv j \pmod{35}.$$

The 12 positions for the set polynomial $g_P(x)$ are

$$P = \{0, 1, 2, 105, 106, 105, 210, 211, 247, 315, 351, 352\}.$$

- The polynomial $f_1(x)$ divides the factor $x^{353} + x^{352} + x^{248} + x^{247} + x^{210} + x^{143} + x$ of $g_P(x)$ and hence folding its sequences into 13×35 arrays does not yield a $(13, 35; 3, 4)$-PRAC.
- The polynomial $f_2(x)$ does not divide $g_P(x)$ and hence folding its sequences into 13×35 arrays yields a $(13, 35; 3, 4)$-PRAC.
- The polynomial $f_3(x)$ divides the factor $x^{353} + x^{352} + x^{247} + x^{210} + x^{106} + x^{105} + x^2 + x$ of $g_P(x)$ and hence folding its sequences into 13×35 arrays does not yield a $(13, 35; 3, 4)$-PRAC.
- The polynomial $f_4(x)$ does not divide $g_P(x)$ and hence folding its sequences into 13×35 arrays yields a $(13, 35; 3, 4)$-PRAC.

As we see, each one of the four irreducible polynomials behaves differently in constructing these two pseudo-random array codes. To summarize the consequences we have

1) Folding the sequences of $f_1(x)$ does not produce a $(13, 35; 4, 3)$-PRAC and also does not produce a $(13, 35; 3, 4)$-PRAC.
2) Folding the sequences of $f_2(x)$ does not produce a $(13, 35; 4, 3)$-PRAC and produces a $(13, 35; 3, 4)$-PRAC.
3) Folding the sequences of $f_3(x)$ produces a $(13, 35; 4, 3)$-PRAC and does not produce a $(13, 35; 3, 4)$-PRAC.
4) Folding the sequences of $f_4(x)$ produces a $(13, 35; 4, 3)$-PRAC and produces a $(13, 35; 3, 4)$-PRAC.

■

VI. PRACs Obtained by Folding

In this section, the second main technique of this paper will be presented. This technique yields a construction for PRACs with various parameters. Infinite families and specific PRACs obtained by the construction will be analysed in the next section.

Definition 14: Let $f_1(x)$ and $f_2(x)$ be two non-constant polynomials. We denote by $f_1(x) \vee f_2(x)$ the polynomial whose roots have the form $\alpha\beta$, where α is a root of $f_1(x)$ and β is a root of $f_2(x)$ in the splitting field of $f_1(x)f_2(x)$.

Example 9: Let $f_1(x) = x^4 + x + 1$ and $f_2(x) = x^3 + x + 1$. The splitting field of $f_1(x)f_2(x)$ is the finite field $\mathbb{F}_{2^{12}}$. Let α be a primitive element of $\mathbb{F}_{2^{12}}$ with the minimal ploynomial $m(x) = x^{12} + x^7 + x^6 + x^5 + x^3 + x + 1$. The roots of $f_1(x)$ are

$$\beta_1 = \alpha^{273}, \beta_2 = \alpha^{546}, \beta_3 = \alpha^{1092}, \beta_4 = \alpha^{2184}.$$

The roots of $f_2(x)$ are

$$\gamma_1 = \alpha^{585}, \gamma_2 = \alpha^{1170}, \gamma_3 = \alpha^{2340}.$$

Finally, $f_1(x) \vee f_2(x) = \prod_{i=1}^{4} \prod_{j=1}^{3} (x - \beta_i\gamma_j) = x^{12} + x^9 + x^5 + x^4 + x^3 + x + 1$. ■

The following proposition, implied by Theorem 8.67 in [20] is a key result for the construction of PRACs. More general results were also proved in [35] and [42]. It shows a relation between vector spaces $S(f_1(x) \vee f_2(x))$ and $S(f_1(x)) \cdot S(f_2(x))$.

Proposition 5: Let $f_1(x)$ and $f_2(x)$ be two polynomials with no repeated roots, and degrees n_1 and n_2, respectively. Moreover, suppose $f_1(x)$ and $f_2(x)$ generate zero factors $\mathrm{ZF}(n_1, r_1)$ and $\mathrm{ZF}(n_2, r_2)$, respectively. Let β_i, $1 \le i \le n_1$ be the distinct roots of $f_1(x)$ and γ_j, $1 \le j \le n_2$ be the distinct roots of $f_2(x)$ in the splitting fields of $f_1(x)f_2(x)$. Then $S(f_1(x)) \cdot S(f_2(x)) = S(f_1(x) \vee f_2(x))$, where

$$f_1(x) \vee f_2(x) = \prod_{i=1}^{n_1} \prod_{j=1}^{n_2} (x - \beta_i\gamma_j).$$

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Proposition 5 implies in other words that the set of sequences generated by $f_1(x) \vee f_2(x)$ is the linear span of the sequences that are obtained as a bitwise product of the sequences generated by $f_1(x)$ and $f_2(x)$. Proposition 5 leads to another construction of PRACs.

Theorem 4: Let r_1, r_2, n_1, n_2 be positive integers such that $\gcd(r_1, r_2) = 1$, and let $f_i(x)$, $i \in \{1, 2\}$, be a non-constant polynomial of degree n_i and uniform exponent r_i. If the feedback shift register with the characteristic polynomial $g(x) \triangleq f_1(x) \vee f_2(x)$ produces a zero factor $\mathrm{ZF}(n_1 n_2, r_1 r_2)$, i.e., $g(x)$ has degree $n_1 n_2$ and uniform exponent $r_1 r_2$, then folding the sequences generated by $g(x)$ into $r_1 \times r_2$ arrays yields an $(r_1, r_2; n_1, n_2)$-PRAC with $(2^{n_1 n_2} - 1)/r_1 r_2$ codewords.

Proof: Let $\mathcal{C}$ be the zero factor obtained from the nonzero sequences of the characteristic polynomial $f_1(x) \vee f_2(x)$. We claim that $\mathcal{F}(\mathcal{C}; r_1, r_2)$ is an $(r_1, r_2; n_1, n_2)$-PRAC with $\ell = (2^{n_1 n_2} - 1)/r_1 r_2$ codewords.

Let $\mathcal{A}$ and $\mathcal{A}'$ be two $r_1 \times r_2$ arrays in $\mathcal{F}(\mathcal{C}; r_1, r_2)$, where either $\mathcal{A}'$ is a non-trivial shift of $\mathcal{A}$, or $\mathcal{A}'$ is different from $\mathcal{A}$. By definition $\mathcal{A} = \mathcal{F}(\boldsymbol{v}; r_1, r_2)$ and $\mathcal{A}' = \mathcal{F}(\boldsymbol{v}'; r_1, r_2)$ for some sequences $\boldsymbol{v}, \boldsymbol{v}' \in \mathcal{C}$, where either $\boldsymbol{v}'$ is a non-trivial shift of $\boldsymbol{v}$, or $\boldsymbol{v}'$ is diffferent from $\boldsymbol{v}$. Since $\mathcal{C}$ is a zero factor generated by an LFSR, it follows from Lemma 2 that $\mathcal{A}+\mathcal{A}' = \mathcal{F}(\boldsymbol{v}; r_1, r_2)+\mathcal{F}(\boldsymbol{v}'; r_1, r_2) = \mathcal{F}(\boldsymbol{v}+\boldsymbol{v}'; r_1, r_2)$. Therefore, $\mathcal{A}+\mathcal{A}'$ is a codeword in $\mathcal{F}(\mathcal{C}; r_1, r_2)$. Since each $n_1 n_2$-tuple appears in exactly one window of a sequence in $\mathcal{C}$ and each sequence of $\mathcal{C}$ has least period $r_1 r_2$ it follows that the number of sequences in $\mathcal{C}$ is $\ell = (2^{n_1 n_2} - 1)/r_1 r_2$ and this is also the number of codewords in $\mathcal{F}(\mathcal{C}; r_1, r_2)$.

By sliding an $n_1 \times n_2$ window over all positions of the ℓ arrays of $\mathcal{F}(\mathcal{C}; r_1, r_2)$, we obtain a set of $2^{n_1 n_2} - 1$ matrices of size $n_1 \times n_2$. It remains to prove that any two such $n_1 \times n_2$ matrices are distinct, and the $n_1 \times n_2$ all-zero matrix is not contained in this set. Note that by Proposition 5, the matrices in $\mathcal{F}(\mathcal{C}; r_1, r_2)$ are spanned by the matrices in $\{\mathcal{F}(\boldsymbol{a} \cdot \boldsymbol{b}, r_1, r_2) \ : \ \boldsymbol{a} \in S(f_1(x)), \ \boldsymbol{b} \in S(f_2(x))\}$. (The fact that the polynomials have a uniform exponent implies their roots are distinct, and so we can apply Proposition 5.) By Lemmas 3, 5, and 6, a codeword $\mathcal{A}$ in $\mathcal{F}(\mathcal{C}; r_1, r_2)$ has the following two properties:

(a) Each column of $\mathcal{A}$ is either all-zero or a nonzero sequence generated by $f_1(x)$.
(b) Each row of $\mathcal{A}$ is either all-zero or a nonzero sequence generated by $f_2(x)$.

Let $\mathcal{A} \in \mathcal{F}(\mathcal{C}; r_1, r_2)$, let X be an arbitrary $n_1 \times n_2$ window from $\mathcal{A}$, and assume first that X is an all-zero matrix. Since each nonzero sequence generated by $f_2(x)$ does not have a run of n_2 consecutive *zeros*, it follows that all the rows of $\mathcal{A}$ that contain X are all-zero rows, i.e., $\mathcal{A}$ contains n_1 consecutive all-zero rows. Therefore, since each nonzero sequence generated by $f_1(x)$ does not have a run of n_1 consecutive *zeros*, it follows that all the columns of $\mathcal{A}$ are all-zero columns, i.e., $\mathcal{A}$ is an all-zero matrix, a contradiction. Thus, the $n_1 \times n_2$ all-zero matrix is not contained in a window of $\mathcal{A}$.

If two $n_1 \times n_2$ windows from $\mathcal{A}, \mathcal{A}' \in \mathcal{F}(\mathcal{C}; r_1, r_2)$ form the same matrix, then their bitwise addition in the related shift forms another codeword with an all-zero $n_1 \times n_2$ window, a contradiction.

Thus, the theorem is proved. ∎

Recall that Proposition 3 exhibits polynomials with a given uniform exponent. Hence, from Lemma 7 and Theorem 4 we have the following consequence.

Corollary 8: Let r_1, r_2, n_1, n_2 be positive integers such that $\gcd(r_1, r_2) = 1$. For $i \in \{1, 2\}$, let $f_i(x)$ be a non-constant polynomial of degree n_i and uniform exponent r_i. Then, the sequences generated by $g(x) \triangleq f_1(x) \vee f_2(x)$ can be folded into $r_1 \times r_2$ arrays and form an $(r_1, r_2; n_1, n_2)$-PRAC.

Example 10: Let $f_1(x) = (x^3 + x^2 + 1)(x^3 + x + 1) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$ be a reducible polynomial with uniform exponent 7 and $f_2(x) = (x^4 + x^3 + 1)(x^4 + x + 1) = x^8 + x^7 + x^5 + x^4 + x^3 + x + 1$ be a reducible polynomial with uniform exponent 15.

The polynomial $f_1(x) \vee f_2(x) = x^{48} + x^{47} + x^{46} + x^{43} + x^{42} + x^{40} + x^{39} + x^{36} + x^{35} + x^{34} + x^{33} + x^{32} + x^{31} + x^{28} + x^{26} + x^{24} + x^{22} + x^{20} + x^{17} + x^{16} + x^{15} + x^{14} + x^{13} + x^{12} + x^9 + x^8 + x^6 + x^5 + x^2 + x + 1 = (x^{12} + x^8 + x^6 + x^5 + x^3 + x^2 + 1)(x^{12} + x^9 + x^5 + x^4 + x^3 + x + 1)(x^{12} + x^{10} + x^9 + x^7 + x^6 + x^4 + 1)(x^{12} + x^{11} + x^9 + x^8 + x^7 + x^3 + 1)$ is a reducible polynomial with uniform exponent 105 which forms a $\mathrm{ZF}(12, 105)$. By Theorem 4 and Corollary 8, folding the sequences of this zero factor into 7×15 arrays yields a $(7, 15; 6, 8)$-PRAC. ∎

VII. Folding Sequences From an Irreducible Polynomial

The main goal of this section is to generalize Theorem 2 for pseudo-random array codes. In the next section we compare the constructed codes with the codes that were constructed based on Theorem 4 and Corollary 8.

Theorem 5: Let n_1, n_2, r_1, r_2 be positive integers such that $r_1 r_2$ divides $2^{n_1 n_2} - 1$ and $\gcd(r_1, r_2) = 1$. If r_1 divides $2^{n_1} - 1$ and the integers $2^i \pmod{r_1}$, $0 \le i \le n_1 - 1$ are distinct, then folding the sequences generated by an irreducible polynomial $g(x)$, of degree $n_1 n_2$ and exponent $r_1 r_2$ into $r_1 \times r_2$ arrays forms an $(r_1, r_2; n_1, n_2)$-PRAC $\mathcal{B}$.

Proof: Let α be a primitive element in $\mathbb{F}_{2^{n_1 n_2}}$, $\ell = \frac{2^{n_1 n_2} - 1}{r_1 r_2}$, and $\delta = \alpha^\ell$. Note that the order of δ is $\eta = r_1 r_2$. Let $g(x)$ be an irreducible polynomial, for which the root of its companion polynomial is δ. Let $\mathcal{S}$ be a nonzero sequence generated by $g(x)$. Furthermore, let G be an $(n_1 n_2) \times (r_1 r_2)$ matrix defined by the $n_1 n_2$ consecutive shifts of $\mathcal{S}$. In such a representation, the columns of G are arranged in the order isomorphic to $\delta^0, \delta^1, \delta^2, \ldots, \delta^{\eta - 1}$ defined by the companion matrix. For (i, j) with $0 \le i < r_1$ and $0 \le j < r_2$, let $\ell(i, j)$ be the integer with $0 \le \ell(i, j) < \eta$ such that $\ell(i, j) \equiv i \pmod{r_1}$ and $\ell(i, j) \equiv j \pmod{r_2}$.

We claim that the $n_1 n_2$ columns of G associated with the elements $\delta^{\ell(i,j)}$, where $\ell(i, j) \equiv i \pmod{r_1}$, $\ell(i, j) \equiv j \pmod{r_2}$, $0 \le i \le n_1 - 1$ and $0 \le j \le n_2 - 1$, are linearly independent. These columns are associated with the $n_1 \times n_2$ sub-array in the upper-left corner of $\mathcal{B}$.

Assume for a contradiction that

$$\sum_{i=0}^{n_1 - 1} \sum_{j=0}^{n_2 - 1} c_{ij} \delta^{\ell(i,j)} = 0,$$

where not all the c_{ij} are zeros. Since $\gcd(r_1, r_2) = 1$, it follows by Bézout's identity that there exist two integers μ and ν such that $\mu r_1 + \nu r_2 = 1$. Let $\beta = \alpha^{\nu r_2}$ and $\gamma = \alpha^{\mu r_1}$. We have

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that $\alpha = \alpha^{\mu r_1 + \nu r_2} = \beta\gamma$. Hence, $\delta^{\ell(i,j)} = (\alpha^\ell)^{\ell(i,j)} = (\beta^i\gamma^j)^\ell$. Therefore, we have

$$\begin{aligned} 0 &= \sum_{i=0}^{n_1-1}\sum_{j=0}^{n_2-1} c_{ij}\delta^{\ell(i,j)} \\ &= \sum_{i=0}^{n_1-1}\sum_{j=0}^{n_2-1} c_{ij}(\beta^i\gamma^j)^\ell \\ &= \sum_{j=0}^{n_2-1}\left(\sum_{i=0}^{n_1-1} c_{ij}(\beta^\ell)^i\right)(\gamma^\ell)^j. \end{aligned} \tag{4}$$

We claim that $\sum_{i=0}^{n_1-1}(c_{i,j}(\beta^\ell)^i) = 0$ for all j.

Since $\beta^{r_1} = 1$ and r_1 divides $2^{n_1}-1$, we have that $\beta \in \mathbb{F}_{2^{n_1}}$ and hence $\sum_{i=0}^{n_1-1} c_{i,j}(\beta^\ell)^i \in \mathbb{F}_{2^{n_1}}$, i.e., the coefficient of $(\gamma^\ell)^j$ in Equation (4) is an element of $\mathbb{F}_{2^{n_1}}$.

Let m be the smallest integer such that $(\gamma^\ell)^m = \alpha^{\mu r_1 \ell m} = \delta^{\mu r_1 m} = 1$. Since the order of δ is r_1r_2, it follows that r_1r_2 divides $\mu r_1 m$. Moreover, $\mu r_1 + \nu r_2 = 1$ and hence $\gcd(\mu, r_2) = 1$ which implies that r_2 divides m, and therefore $r_2 \le m$. However, $(\gamma^\ell)^{r_2} = \alpha^{\mu r_1 \ell r_2} = \delta^{\mu r_1 r_2} = 1$ and since m is the smallest integer for which $(\gamma^\ell)^m = 1$, it follows that $m = r_2$ which implies that γ^ℓ has order r_2. We claim that

$$\gamma^\ell, (\gamma^\ell)^{2^{n_1}}, (\gamma^\ell)^{2^{2n_1}}, \ldots, (\gamma^\ell)^{2^{(n_2-1)n_1}}$$

are all distinct.

To see this, suppose for a contradiction that they are not distinct. Then there exists $1 \le e \le n_2 - 1$ such that $(\gamma^\ell)^{2^{n_1 e}-1} = 1$. Since $\gamma^\ell = \alpha^{\mu r_1 \ell} = \delta^{\mu r_1}$, $\gcd(r_2, \mu) = 1$, and the order of δ is r_1r_2, it follows that r_1r_2 divides $\mu r_1(2^{n_1 e}-1)$, so r_2 divides $2^{n_1 e}-1$. Let m be the smallest integer such that $\delta^{2^m-1} = 1$. Since the order of δ is r_1r_2 we have that r_1r_2 divides $2^m - 1$. Since r_2 divides $2^{n_1 e}-1$ and r_1 divides $2^{n_1}-1$, it follows that r_1 divides $2^{n_1 e}-1$ and hence $2^{n_1 e} - 1 = r_1 r_2 x$, for a positive integer x. Therefore, $m \le n_1 e$ and hence $g(x)$ has fewer than n_1n_2 distinct roots in $\mathbb{F}_{2^{n_1 n_2}}$, a contradiction. Hence, the claim follows. This implies that

$$\gamma^\ell, (\gamma^\ell)^{2^{n_1}}, (\gamma^\ell)^{2^{2n_1}}, \ldots, (\gamma^\ell)^{2^{(n_2-1)n_1}}$$

are distinct, and $(\gamma^\ell)^{2^{n_1 n_2}} = \gamma^\ell$. Since $(\gamma^\ell)^{2^{n_1 n_2}} = \gamma^\ell$, it follows that

$$\prod_{i=0}^{n_2-1}(x - (\gamma^\ell)^{2^{in_1}})$$

is the minimal polynomial of γ^ℓ in $\mathbb{F}_{2^{n_1}}$. This polynomial has degree n_2 in $\mathbb{F}_{2^{n_1}}$. However, the polynomial in Equation (4) is a polynomial in γ^ℓ of smaller degree $n_2 - 1$ which is equal to zero. This implies that all the coefficients of $(\gamma^\ell)^j$ in Equation (4) are equal to zero. Therefore, for each $0 \le j \le n_2 - 1$ we have

$$\sum_{i=0}^{n_1-1} c_{ij}(\beta^\ell)^i = 0.$$

Similarly we can prove that r_1 is the smallest positive integer m such that $(\beta^\ell)^m = 1$, i.e., β^ℓ has order r_1. Since the n_1 integers $2^i \pmod{r_1}$ for $0 \le i \le n_1 - 1$ are distinct, we have that

$$\beta^\ell, (\beta^\ell)^2, (\beta^\ell)^{2^2}, \ldots, (\beta^\ell)^{2^{n_1-1}}$$

are distinct, and $(\beta^\ell)^{2^{n_1 n_2}} = \beta^\ell$. Therefore, the minimal polynomial of β^ℓ over $\mathbb{F}_2$ has degree n_1 which implies that $c_{ij} = 0$ for $0 \le i \le n_1 - 1$ and $0 \le j \le n_2 - 1$. So we have a contradiction, as required.

Therefore, the $n_1 \times n_2$ matrix in the upper-left corner of $\mathcal{B}$ is a nonzero matrix and folding the sequences generated by $g(x)$ yields an $(r_1, r_2; n_1, n_2)$-PRAC. ■

Remark: The three proofs (Theorem 1 proved in [22], Theorem 2, and Theorem 5) for the correctness of the constructions of PRAs using folding are very similar, but at various points in the proof the arguments are different, because of the different conditions required for the construction.

Example 11: Let $n_1 = 3$, $n_2 = 4$, $r_1 = 7$, and $r_2 = 13$ and consider the irreducible polynomial $g(x) = x^{12}+x^{10}+x^9+x+1$ with degree 12 and exponent 91. This polynomial, r_i and n_i, $i = 1, 2$, satisfy the conditions of Theorem 5 (because $7 \cdot 13 = 91$ divides $2^{3\cdot 4} - 1$, $\gcd(7, 13) = 1$, 7 divides $2^3 - 1$, and $2^0 - 1$, $2^1 = 2$, $2^2 = 4$ are all distinct modulo 7) and hence folding the sequences generated by $g(x)$ into 7×13 arrays yields a $(7, 13; 3, 4)$-PRAC.

There are 6 different irreducible polynomials of degree 12 and exponent 91, each one of them yields a $(7, 13; 3, 4)$-PRAC. ■

TABLE I
TYPES OF PRACS

$f_1(x)$	$f_2(x)$	$g(x)$	references
reducible	reducible	reducible	Lemma 29, Example 10
reducible	INP	reducible	Lemma 29, Example 12
reducible	primitive	reducible	Lemma 29, Example 13
INP	INP	reducible	Corollary 32, Example 14
INP	INP	INP	Corollary 33, Example 16
INP	primitive	reducible	Corollary 32, Example 15
INP	primitive	INP	Corollary 33, Example 17
primitive	primitive	INP	Corollary 33, Example 18

VIII. Analysis of the Constructions

We divide polynomials into three ***types***: primitive, INP, and reducible. In this section, the construction of PRACs from Section VI, by folding the sequences generated by $g(x) = f_1(x) \vee f_2(x)$, is investigated. Which combinations of types of the polynomials $f_1(x)$, $f_2(x)$ and $g(x)$ are possible? The results of the section are summarized in Table I at the end of the section: we provide examples of all combinations that do occur, and show why all other combinations cannot occur. We also write down several families of PRACs which arise from the construction in Section VI.

A. Some Impossible Types

To understand which combinations of these three types of polynomials are possible we will have to use Proposition 4 and Corollary 4. These results enable us to decide which combinations of the three types of polynomials for $f_1(x)$, $f_2(x)$, and $g(x)$ are not possible.

Lemma 10: The polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 4 and Corollary 8 cannot be a primitive polynomial.

Proof: If $g(x)$ is a primitive polynomial, then it generates exactly one nonzero sequence. Let $f_i(x)$, $i = 1, 2$, be a

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polynomial of degree n_i and exponent r_i. The number of nonzero sequences generated by $f_i(x)$ is $\frac{2^{n_i}-1}{r_i}$ which is at least 1. The degree of $g(x)$ is n_1n_2, and its exponent is r_1r_2. The number of nonzero sequences generated by $g(x)$ is

$$\frac{2^{n_1n_2}-1}{r_1r_2} > \frac{2^{n_1}-1}{r_1}\frac{2^{n_2}-1}{r_2} \geq 1 \ .$$

Therefore, $g(x)$ is not a primitive polynomial. ∎

While Lemma 10 asserts that $g(x)$ cannot be a primitive polynomial, the next lemma asserts that $g(x)$ is a reducible polynomial if $f_1(x)$ or $f_2(x)$ is a reducible polynomial

Lemma 11: If $f_1(x)$ or $f_2(x)$ is a reducible polynomial, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 4, is a reducible polynomial.

Proof: Let $f_i(x)$, $i = 1, 2$, be a polynomial of degree n_i and exponent r_i. The polynomial $g(x) = f_1(x) \vee f_2(x)$ has degree n_1n_2 and exponent r_1r_2. The nonzero sequences generated by $g(x)$ have least period r_1r_2 and they are folded into $r_1 \times r_2$ arrays to form an $(r_1, r_2; n_1, n_2)$-PRAC with $\frac{2^{n_1n_2}-1}{r_1r_2}$ codewords. Without loss of generality, assume that $f_1(x)$ is a reducible polynomial. By Corollary 4, there exists an irreducible polynomial $f_3(x)$ with exponent r_1 which is a proper factor of $f_1(x)$ and whose degree n_3 is smaller than n_1. The polynomial $f_3(x) \vee f_2(x)$ also has exponent r_1r_2, it divides $g(x)$ but its degree n_3n_2 is smaller than n_1n_2, the degree of $g(x)$, and hence again by using Corollary 4 we have that $g(x)$ is a reducible polynomial. ∎

Example 12: Let $f_1(x) = \sum_{i=1}^{6} x^i + 1 = (x^3 + x + 1)(x^3 + x^2 + 1)$ be a reducible polynomial with exponent 7 and $f_2(x) = \sum_{i=1}^{10} x^i + 1$ be an INP polynomial with exponent 11. $f_1(x) \vee f_2(x) = (x^{30} + x^{28} + x^{27} + x^{26} + x^{23} + x^{21} + x^{20} + x^{19} + x^{16} + x^{14} + x^{13} + x^{12} + x^9 + x^8 + x^7 + x^4 + x^2 + x + 1)(x^{30} + x^{29} + x^{28} + x^{26} + x^{23} + x^{22} + x^{21} + x^{18} + x^{17} + x^{16} + x^{14} + x^{11} + x^{10} + x^9 + x^7 + x^4 + x^3 + x^2 + 1)$. These two factors of $f_1(x) \vee f_2(x)$ are INP polynomials with exponent $77 = 7 \cdot 11$ and therefore $f_1(x) \vee f_2(x)$ forms a ZF(60, 77). By Theorem 4 and Corollary 8, folding its sequences into 7×11 arrays produces a $(7, 11; 6, 10)$-PRAC. ∎

Example 13: Let $f_1(x) = \sum_{i=0}^{6} x^i = (x^3 + x^2 + 1)(x^3 + x + 1)$ be a reducible polynomial with exponent 7 and let $f_2(x) = x^2 + x + 1$ be a primitive polynomial with exponent 3. The polynomial

$$f_1(x) \vee f_2(x) = (x^6 + x^4 + x^2 + x + 1)(x^6 + x^5 + x^4 + x^2 + 1)$$

is a reducible polynomial with exponent 21. By Theorem 4 and Corollary 8, folding its sequences into 7×3 arrays produces a $(7, 3; 6, 2)$-PRAC. ∎

B. Some Possible Types

For the next combinations of polynomial types for $f_1(x)$, $f_2(x)$ and $g(x)$, the following well-known lemma is required.

Lemma 12: Let r_i and n_i, $i = 1, 2$, be positive integers such that $\gcd(r_1, r_2) = 1$ and

$$2^{n_i} \equiv 1 \pmod{r_i}. \tag{5}$$

If $\ell = \gcd(n_1, n_2)$ then

$$2^{n_1n_2/\ell} \equiv 1 \pmod{r_1r_2}.$$

If n_1 and n_2 are the smallest integers that satisfy Equation (5), then n_1n_2/ℓ is the smallest value of k which satisfies $2^k \equiv 1 \pmod{r_1r_2}$.

Lemma 13: If $f_i(x)$, $i = 1, 2$, are polynomials of degree n_i and exponent r_i such that $\gcd(r_1, r_2) = 1$ and $\gcd(n_1, n_2) > 1$, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 4 is a reducible polynomial.

Proof: Let $f_i(x)$, $i = 1, 2$, be polynomials of degree n_i and exponent r_i. The polynomial $g(x) = f_1(x) \vee f_2(x)$ has degree n_1n_2 and exponent r_1r_2. The nonzero sequences generated by $g(x)$ have least period r_1r_2 and they are folded into $r_1 \times r_2$ arrays to form an $(r_1, r_2; n_1, n_2)$-PRAC with $\frac{2^{n_1n_2}-1}{r_1r_2}$ codewords. By Lemma 11 if $f_1(x)$ or $f_2(x)$ is reducible, then $g(x)$ is a reducible polynomial and the lemma follows. Hence, we may assume that both $f_1(x)$ and $f_2(x)$ are irreducible polynomials. By Proposition 4 $2^{n_1} \equiv 1 \pmod{r_1}$ and $2^{n_2} \equiv 1 \pmod{r_2}$. By Proposition 4 if $g(x)$ is an irreducible polynomial, then

$$2^{n_1n_2} \equiv 1 \pmod{r_1r_2} \tag{6}$$

and n_1n_2 is the multiplicative order of 2 modulo r_1r_2. But, since $\ell = \gcd(n_1, n_2) > 1$, it follows from Lemma 12 that $2^{n_1n_2/\ell} \equiv 1 \pmod{r_1r_2}$ and hence n_1n_2 is not the multiplicative order of 2 modulo r_1r_2. Therefore, $g(x)$ is not an irreducible polynomial, i.e., it is a reducible polynomial. ∎

Corollary 9: If $f_i(x)$, $i = 1, 2$ are polynomials of degree n_i with exponent r_i, such that $\gcd(n_1, n_2) > 1$ and $f_1(x)$ or $f_2(x)$ is an INP polynomial, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 4 and Corollary 8 is a reducible polynomial.

Example 14: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an INP polynomial with exponent 5 and $f_2(x) = x^6 + x^3 + 1$ be an INP polynomial with exponent 9.

The polynomial $f_1(x) \vee f_2(x) = x^{24} + x^{21} + x^{15} + x^{12} + x^9 + x^3 + 1 = (x^{12} + x^9 + 1)(x^{12} + x^3 + 1)$ is a reducible polynomial with exponent 45 which forms a ZF(24, 45). By Theorem 4 and Corollary 8, folding the sequences of this zero factor into 5×9 arrays produces a $(5, 9; 4, 6)$-PRAC. ∎

Example 15: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an INP polynomial with exponent 5 and $f_2(x) = x^6 + x^5 + 1$ be a primitive polynomial with exponent 63.

The polynomial $f_1(x) \vee f_2(x) = (x^{12} + x^4 + x^2 + x + 1)(x^{12} + x^{11} + x^{10} + x^9 + x^8 + x^6 + x^3 + x + 1)$. has two factors. These two factors of $f_1(x) \vee f_2(x)$ are irreducible polynomials with exponent $315 = 5 \cdot 63$, and therefore $f_1(x) \vee f_2(x)$ forms a ZF(24, 315). By Theorem 4 and Corollary 8, folding its sequences into 5×63 arrays produces a $(5, 63; 4, 6)$-PRAC. ∎

Corollary 10: If $f_i(x)$, $i = 1, 2$, are irreducible polynomials of degree n_i and exponent r_i such that $\gcd(n_1, n_2) = 1$, then the polynomial $g(x) = f_1(x) \vee f_2(x)$ in Theorem 4 and Corollary 8, is an INP polynomial.

Proof: Since $f_i(x)$ is an irreducible polynomial, it follows from Proposition 4 that $2^{n_i} \equiv 1 \pmod{r_i}$ and n_i is the smallest integer that satisfies this equation. By Lemma 12 we have that $k = n_1n_2$ is the smallest integer that satisfies the equation

$$2^k \equiv 1 \pmod{r_1r_2}. \tag{7}$$

Assume for a contradiction that $g(x)$ is a reducible polynomial of degree n_1n_2 and exponent r_1r_2. It follows by Corollary 4 that there exists an irreducible polynomial $h(x)$ which divides $g(x)$

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whose degree n_3 divides n_1n_2 and its exponent is r_1r_2, Hence there exists an integer smaller than n_1n_2 which is a solution for k in Equation (7), which is a contradiction. Therefore, $g(x)$ is an irreducible polynomial, and by Lemma 10 it cannot be a primitive polynomial as required. ∎

Example 16: Let $f_1(x) = x^4 + x^3 + x^2 + x + 1$ be an INP polynomial with exponent 5 and $f_2(x) = x^9 + x + 1$ be an INP polynomial with exponent 73.

The polynomial $f_1(x) \vee f_2(x) = x^{36} + x^{28} + x^{27} + x^{20} + x^{18} + x^{12} + x^{10} + x^9 + x^4 + x^3 + x^2 + x + 1$ is an INP polynomial with exponent $365 = 5 \cdot 73$. By Theorem 4 and Corollary 8, folding its sequences into 5×73 arrays yields a $(5, 73; 4, 9)$-PRAC.∎

Example 17: Let $f_1(x) = \sum_{i=1}^{4} x^i$ be an INP polynomial with exponent 5 and Let $f_2(x) = x^3 + x^2 + 1$ be a primitive polynomial with exponent 7. The polynomial

$$f_1(x) \vee f_2(x) = x^{12} + x^{11} + x^{10} + x^8 + x^5 + x^4 + x^3 + x^2 + 1$$

is an INP polynomial with exponent $35 = 7 \cdot 5$. By Theorem 4 and Corollary 8, folding its sequences into 5×7 arrays yields a $(5, 7; 4, 3)$-PRAC. ∎

C. The Number of Irreducible Polynomials of Fixed Exponent

How many irreducible polynomials of degree n and exponent e exist? Some information on this question is given in the following results. The first result can be found in [14, p. 41].

Lemma 14: If e is a factor of $2^n - 1$ which is not a factor of any number $2^\ell - 1$, where $\ell < n$, then there are exactly $\frac{\phi(e)}{n}$ irreducible polynomials with exponent e, where ϕ is the Euler totient function.

Lemma 15: If there are k_i primitive polynomials of degree n_i, $i = 1, 2$, and $\gcd(2^{n_1} - 1, 2^{n_2} - 1) = 1$, then there are k_1k_2 irreducible polynomials with exponent $(2^{n_1} - 1)(2^{n_2} - 1)$.

Proof: By Lemma 14 there are $k_1 = \frac{\phi(2^{n_1}-1)}{n_1}$ primitive polynomials of degree n_1 and exponent $2^{n_1} - 1$. By Lemma 14 there are $k_2 = \frac{\phi(2^{n_2}-1)}{n_2}$ primitive polynomials of degree n_2 and exponent $2^{n_2} - 1$. By Lemma 14 there are $\frac{\phi((2^{n_1}-1)(2^{n_2}-1))}{n_1n_2}$ irreducible polynomials of degree n_1n_2 and exponent $(2^{n_1} - 1)(2^{n_2} - 1)$. Now,

$$\frac{\phi((2^{n_1} - 1)(2^{n_2} - 1))}{n_1n_2} = \frac{\phi(2^{n_1} - 1)}{n_1} \frac{\phi(2^{n_2} - 1)}{n_2} = k_1k_2.$$

∎

Lemma 15 implies each polynomial of degree n_1n_2 and exponent $(2^{n_1} - 1)(2^{n_2} - 1)$ is obtained from two distinct primitive polynomials of degrees n_1 and n_2 via $f_1(x) \vee f_2(x)$.

Example 18: Consider primitive polynomials $f_1(x)$ and $f_2(x)$ of degree 3 and 4, respectively, with exponents 7 and 15, respectively.

If $f_1(x) = x^4 + x + 1$ and $f_2(x) = x^3 + x + 1$ then

$$f_1(x) \vee f_2(x) = x^{12} + x^9 + x^5 + x^4 + x^3 + x + 1.$$

If $f_1(x) = x^4 + x + 1$ and $f_2(x) = x^3 + x^2 + 1$ then

$$f_1(x) \vee f_2(x) = x^{12} + x^8 + x^6 + x^5 + x^3 + x^2 + 1.$$

If $f_1(x) = x^4 + x^3 + 1$ and $f_2(x) = x^3 + x + 1$ then

$$f_1(x) \vee f_2(x) = x^{12} + x^{10} + x^9 + x^7 + x^6 + x^4 + 1.$$

If $f_1(x) = x^4 + x^3 + 1$ and $f_2(x) = x^3 + x^2 + 1$ then

$$f_1(x) \vee f_2(x) = x^{12} + x^{11} + x^9 + x^8 + x^7 + x^3 + 1.$$

In each of these four cases, $f_1(x) \vee f_2(x)$ is an INP polynomial with exponent $105 = 15 \cdot 7$ that forms a ZF$(12, 105)$. By Theorem 4 and Corollary 8, folding the sequences of this zero factor into 15×7 arrays is an $(15, 7; 4, 3)$-PRAC. ∎

Lemma 15 can be generalized as follows.

Lemma 16: Suppose there exist k_i irreducible polynomials of degree n_i with exponent r_i, $i = 1, 2$. Assume that $\gcd(r_1, r_2) = 1$ and $\gcd(n_1, n_2) = \ell$. Then there are exactly ℓk_1k_2 irreducible polynomials of degree $\frac{n_1n_2}{\ell}$ and exponent r_1r_2. These polynomials can be partitioned into k_1k_2 sets of size ℓ. Each such set is associated with exactly one pair of polynomials $(f_1(x), f_2(x))$, such that f_i has degree n_i, $i = 1, 2$, and $f_1(x) \vee f_2(x)$ is the product of the polynomials in the set.

Proof: Since n_i is the smallest integer such that $2^{n_i} \equiv 1 \bmod r_i$ for $i = 1, 2$ and $\gcd(r_1, r_2) = 1$, by Lemma 12, we know that $n = \frac{n_1n_2}{\ell}$ is the smallest integer such that $2^n \equiv 1 \bmod r_1r_2$. Then by Lemma 14, we know that there are $\frac{\phi(r_1r_2)}{n}$ irreducible polynomials with exponent r_1r_2. In addition, by Proposition 4, all these irreducible polynomials with exponents r_1r_2 have degree n. Therefore, there are $\frac{\phi(r_1r_2)}{n}$ irreducible polynomials with exponent r_1r_2 and degree n. Now it remains to show that $\frac{\phi(r_1r_2)}{n} = \ell k_1k_2$. Note that by Lemma 14, $\phi(r_i) = n_i \cdot k_i$ for $i = 1, 2$. We have $\frac{\phi(r_1r_2)}{n} = \frac{\phi(r_1)\phi(r_2)}{n} = \frac{n_1k_1n_2k_2}{n_1n_2/\ell} = \ell k_1k_2$ and the first claim follows.

Let P be the set of all these irreducible polynomials with exponent r_1r_2 and degree n, and R be their roots in $\mathbb{F}_{2^n}$. Note that R consists of all elements in $\mathbb{F}_{2^n}$ with order r_1r_2. Let P_i be the sets of the k_i irreducible polynomials with degree n_i and uniform exponent r_i in $\mathbb{F}_{2^{n_i}}$, and R_i be their roots, where $i = 1, 2$.

To show the second claim, we first prove that there is a bijective mapping f from R to $R_1 \times R_2$, that is every $\gamma \in R$ has the unique pair $(\alpha, \beta) \in R_1 \times R_2$ with $r = \alpha \cdot \beta$. Since $\gcd(r_1, r_2) = 1$, there exist integers x and y such that $xr_1 + yr_2 = 1$. Let $\alpha = \gamma^{yr_2}$ and $\beta = \gamma^{xr_1}$. Then $\gamma = \alpha \cdot \beta$. In addition, since the order of γ is r_1r_2, $\gcd(y, r_1) = 1$, and $\gcd(x, r_2) = 1$, we have $\alpha \in R_1$ and $\beta \in R_2$. We can easily check that this mapping is injective. Then $|f(R)| = |R| = \ell k_1k_2 \cdot n = k_1n_1 \cdot k_2n_2 = |R_1 \times R_2|$, shows that f is a bijection.

Recall that $f_1(x) \vee f_2(x) = \prod_{\alpha \in R(f_1), \beta \in R(f_2)} (x - \alpha\beta)$, where $R(g)$ denotes the roots of a polynomial g. Then each root γ of any $f_1(x) \vee f_2(x)$ is associated with a pair $(\alpha, \beta) \in R_1 \times R_2$ by $\gamma = \alpha \cdot \beta$. Such pairs derived from the roots of all $f_1(x) \vee f_2(x)$ are distinct since the roots of a polynomial in $P_1(P_2)$ are distinct and the roots of distinct irreducible polynomials are disjoint. Actually, the set of such pairs is equal to $R_1 \times R_2$. From the previous paragraph, we know that the roots of all $f_1(x) \vee f_2(x)$ are in R, which means that

(1) all the roots are distinct,
(2) each root of $f_1(x) \vee f_2(x)$ is also a root of a polynomial in P.

From (2), we know that $f_1(x) \vee f_2(x)$ is a product of some polynomials in P and (1) implies that these polynomials are distinct. Since each factor has $\frac{n_1n_2}{\ell}$ distinct roots and $f_1(x) \vee f_2(x)$ has n_1n_2 distinct roots, it follows that there are ℓ polynomial factors. This implies that each $f_1(x) \vee f_2(x)$

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is a product of ℓ distinct polynomials in P. By (1), the k_1k_2 sets of irreducible factors for each $f_1(x) \vee f_2(x)$ are disjoint. The number of irreducible factors from all sets is $k_1k_2 \cdot \ell = |P|$, which means that the sets form a partition of P which concludes the proof. ■

D. A Summary of Types

There are no more possible combinations of primitive, irreducible non-primitive, and reducible polynomials for $f_1(x)$, $f_2(x)$, and $g(x)$, beside those which are demonstrated in the results and examples of this section. These possible combinations are summarized in Table I. The eight categories of the possible combinations partition the PRACs which were constructed via Theorem 4 and Corollary 8 into equivalence classes. They form a hierarchy between the various PRACs constructed so far.

E. Some Infinite Families of Parameters

We complete this part of the section with some specific infinite families of parameters for which there exist PRACs. This list is only a drop in the ocean of the possible parameters for such codes. These parameters were chosen since they look attractive to the authors. However, many other sets of infinite families of parameters can be found relatively easy.

Corollary 11: If n_1 and n_2 are two positive integers such that $\gcd(2^{n_1}-1, 2^{n_2}-1) = 1$, then there exists a $(2^{n_1}-1, 2^{n_2}-1; n_1, n_2)$-PRAC.

Proof: The claim follows from the fact that there exist M-sequences of length $2^{n_1}-1$ and length $2^{n_2}-1$. This implies that the conditions of Theorem 4 and Corollary 8 are satisfied and the claim follows. ■

Corollary 12: If n_1 and n_2 are two positive integers such that $\gcd(2^{n_1}-1, 2^{n_2}-1) = 1$ and there exist k primitive polynomials of degree n_1 and ℓ primitive polynomials of degree n_2, then there exists a $(2^{n_1}-1, 2^{n_2}-1; kn_1, \ell n_2)$-PRAC.

Corollary 13: If n_1+1 and n_2+1 are distinct primes, then there exists an $(n_1+1, n_2+1; n_1, n_2)$-PRAC.

Proof: If p is a prime, then the polynomial $\sum_{i=0}^{p-1} x^i$ has exponent p and hence it forms a $\mathrm{ZF}(p-1, p)$. By definition $\gcd(n_1+1, n_2+1) = 1$ which implies that the conditions of Theorem 4 and Corollary 8 are satisfied and the claim follows. ■

Corollary 14: If n_1 and n_2 are two positive integers such that $\gcd(2^{n_1}-1, n_2+1) = 1$ and n_2+1 is a prime, then there exists a $(2^{n_1}-1, n_2+1; n_1, n_2)$-PRAC.

F. Another Hierarchy of PRACs

The following lemma yields a second hierarchy between the PRACs generated by the multiplication of irreducible polynomials of the same degree with the same exponent. It is an immediate consequence of the discussion in Lemma 7, Theorem 4, and Corollary 8.

Lemma 17: Let $f_1(x)$ be an irreducible polynomial of degree n_1 and exponent r_1 and let $h_i(x)$, $1 \le i \le \ell$, $\ell \ge 2$ be ℓ irreducible polynomials of degree n_2 and exponent r_2. The arrays of the $(r_1, r_2; n_1, \ell n_2)$-PRAC generated by folding the sequences from $f_1(x) \vee \prod_{i=1}^{\ell} h_i(x)$ into $r_1 \times r_2$ arrays contain the arrays of the $(r_1, r_2; n_1, (\ell-1)n_2)$-PRAC generated by folding the sequences from $f_1(x) \vee \prod_{i=1}^{\ell-1} h_i(x)$ into $r_1 \times r_2$ arrays.

Unfortunately, Lemma 17 does not give a complete picture of the hierarchy. For example, there are eight irreducible polynomials of degree 8 and with exponent 85. Folding the sequences generated by one of these polynomials yields a $(5, 17; 4, 2)$-PRAC. This PRAC is covered by Theorem 5. Folding the sequences of the multiplication of two such polynomials yields a $(5, 17; 4, 4)$-PRAC. This PRAC is not covered by any theorem that was proved here.

Conjecture 1: Suppose that $n_1 < r_1 < 2n_1$, and there are ℓ irreducible polynomials $f_1(x), f_2(x) \ldots f_\ell(x)$ of degree n_1n_2 and exponent r_1r_2. Suppose the set of sequences generated by each $f_i(x)$ forms an $(r_1, r_2; n_1, n_2)$-PRAC. Folding the sequences generated by a product of k such polynomials, $1 \le k \le \ell$, yields an $(r_1, r_2; n_1, k\ n_2)$-PRAC.

The condition $n_1 < r_1 < 2n_1$ in Conjecture 1 is necessary as demonstrated in the next example.

Example 19: Consider the two irreducible polynomials of degree 6 and exponent 21, $f_1(x) = x^6 + x^5 + x^4 + x^2 + 1$ and $f_2(x) = x^6 + x^4 + x^2 + x + 1$. The set of sequences generated by each one yields a $(3, 7; 2, 3)$-PRAC. Folding the product of these two polynomials yield a $(3, 7; 2, 6)$-PRAC.

The set of sequences generated by each one also yields a $(7, 3; 3, 2)$-PRAC, but now the condition $n_1 < r_1 < 2n_1$ is not satisfied with the product of these two polynomials.

There are six primitive polynomials of degree 6 and exponent 63. The sequence generated by each one yields a $(7, 9; 3, 2)$-PRAC. The condition $n_1 < r_1 < 2n_1$ is not satisfied for the product of these polynomials. Consider the two primitive polynomials $f_1(x) = x^6+x^5+1$ and $f_2(x) = x^6+x+1$. Folding the product of these two polynomials does not yield a $(7, 9; 3, 4)$-PRAC. ■

G. Comparing PRAC Constructions

Finally, it is interesting to compare the PRACs constructed in Section VI, which were extensively studied in the current section, and with the PRACs constructed in Section VII.

Lemma 18: The construction of PRACs using Theorem 4 and Corollary 8, where the folded sequences were generated by an irreducible polynomial are generated also by Theorem 5.

Proof: The conditions in Theorem 4 and Corollary 8 are that the sequences that are folded into an $r_1 \times r_2$ array are generated by an irreducible polynomial $g(x)$ whose degree is n_1n_2 and its exponent is r_1r_2 and the polynomial $f_1(x)$ has degree n_1 and exponent r_1. By Table I we have that $f_1(x)$ is an irreducible polynomial. By Proposition 4 we have that n_1 is the smallest integer such that $2^{n_1} \equiv 1 \pmod{r_1}$. This implies that r_1 divides $2^{n_1}-1$ and the integers $2^i \pmod{r_1}$, $0 \le i \le n_1-1$ are distinct which are exactly the required conditions in Theorem 5. ■

The PRACs produced using the construction implied by Theorem 4 and Corollary 8 in which $g(x)$ is a reducible polynomial cannot be constructed via Theorem 5, since the sequences that are folded in Theorem 5 must be generated by an irreducible polynomial. On the other hand, all other PRACs produced via Theorem 4 and Corollary 8 are also generated via Theorem 5 as proved in Lemma 18. All the $(r_1, r_2; n_1, n_2)$-PRACs for which r_1 divides $2^{n_1}-1$ and r_2 does not divide $2^{n_2}-1$, which are generated via Theorem 5 (see Example 11),

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cannot be generated via Theorem 4 and Corollary 8 since the definition of the polynomials $f_1(x)$ and $f_2(x)$ requires these divisibility conditions when $g(x)$ is an irreducible polynomial.

H. Small Parameter Sets

There are 9 possible parameters for $(r_1, r_2; n_1, n_2)$-PRACs, where $r_1 r_2 \leq 1023$. A $(3, 7; 2, 3)$-PRAC of size 3 can be constructed via both Theorems 4 and 5. A $(3, 17; 2, 4)$-PRAC of size 5 and a $(3, 5; 2, 4)$-PRAC of size 17 can be constructed by Theorem 5. A $(5, 17; 2, 4)$-PRAC of size 3 and a $(5, 17; 4, 2)$-PRAC of size 3 obtained by folding the sequences generated by an irreducible polynomial with exponent 85 exists, but cannot be obtained by our theorems. A $(3, 31; 2, 5)$-PRAC of size 11 can be constructed via both Theorem 4 and 5. A $(11, 31; 2, 5)$-PRAC of size 3 and A $(3, 11; 2, 5)$-PRAC of size 31 can be constructed by Theorem 5. A $(11, 31; 5, 2)$-PRAC of size 3 constructed by folding does not exist.

IX. A Necessary and Sufficient Condition to Form a PRAC

In Section V we gave a necessary and sufficient condition to guarantee that folding a sequence or sequences generated by an irreducible polynomial yields a PRA or a PRAC, respectively. In this section, we provide a different necessary and sufficient condition that also works on sequences generated from reducible polynomials. This approach is more computationally efficient for larger parameters than our earlier approach.

A. Checking If an Irreducible Polynomial Yields a PRAC

Let $f_1(x), f_2(x), \ldots, f_k(x)$ be distinct irreducible polynomials over $\mathbb{F}_2$ with degree n and exponent e. Let $f(x) = \prod_{u=1}^{k} f_u(x)$. We are interested in finding a criterion to determine whether the folding of a set of sequences with characteristic polynomial $f(x)$ produces a PRAC (or a PRA if the code contains exactly one codeword). The criterion we provide can be checked in quadratic time, by finding the determinant of a certain binary matrix. We represent the sequences with characteristic polynomial $f(x)$ in the following way:

Lemma 19: Let $f_1(x), f_2(x), \ldots, f_k(x)$ be distinct irreducible polynomials over $\mathbb{F}_2$ of degree n. Let $f(x) = \prod_{u=1}^{k} f_u(x)$. For $1 \leq u \leq k$, let $\alpha_u \in \mathbb{F}_{2^n}$ be a root of $f_u(x)$. Let $t_1, t_2, \ldots, t_k : \mathbb{F}_{2^n} \to \mathbb{F}_2$ be non-trivial $\mathbb{F}_2$-linear maps. Let (s_i) be a sequence with characteristic polynomial $f(x)$. Then there are unique elements $\sigma_1, \sigma_2, \ldots, \sigma_k \in \mathbb{F}_{2^n}$ such that the terms of (s_i) can be written in the form

$$s_i = \sum_{u=1}^{k} t_u(\sigma_u \alpha_u^i) \text{ for } i \geq 0. \tag{8}$$

We follow the approach in Rueppel [31, Section 3.3]), using the trace map. Recall that the *trace map* from $\mathbb{F}_{2^n}$ to $\mathbb{F}_2$ is the function $\mathrm{Tr} : \mathbb{F}_{2^n} \to \mathbb{F}_2$ by $\mathrm{Tr}(x) = x + x^2 + x^{2^2} + \cdots x^{2^{n-1}}$. The basic properties of this map are covered in Lidl and Niederreiter [20, Section 2.3] (where the notation $\mathrm{Tr}_{\mathbb{F}_{2^n}/\mathbb{F}_2}$ is used for this map).

Proof: Let (s_i) be a sequence with characteristic polynomial $f(x)$. Because $f_1(x), f_2(x), \ldots, f_k(x)$ are coprime polynomials, we may write

$$(s_i) = \sum_{u=1}^{k} (s_i^u)$$

for unique sequences (s_i^u) where (s_i^u) has characteristic polynomial $f_u(x)$. (This is a consequence of the properties of partial fractions, when we represent sequences as rational functions. See [20, Theorem 6.55] and the discussion following it, for example.) By [20, Theorem 6.24], there exist unique elements $\theta_1, \theta_2, \ldots, \theta_k \in \mathbb{F}_{2^n}$ such that $s_i^u = \mathrm{Tr}(\theta_u \alpha_u^i)$ for $i \geq 0$. Thus

$$s_i = \sum_{u=1}^{k} \mathrm{Tr}(\theta_u \alpha_u^i) \tag{9}$$

for unique elements $\theta_1, \theta_2, \ldots, \theta_k \in \mathbb{F}_{2^n}$.

By [20, Theorem 2.24], we have that there are unique elements $\tau_1, \tau_2, \ldots, \tau_k \in \mathbb{F}_{2^n}$ such that $t_u(z) = \mathrm{Tr}(\tau_u z)$ for $1 \leq u \leq k$ and all $z \in \mathbb{F}_{2^n}$. The elements τ_u are non-zero, as the maps t_u are non-trivial. So (9) implies that (8) holds for unique elements $\sigma_1, \sigma_2, \ldots, \sigma_k \in \mathbb{F}_{2^n}$, where $\sigma_u = \tau_u \theta_u$ for $1 \leq u \leq k$. ∎

Let r_1 and r_2 be coprime positive integers. Let μ and ν be integers such that $\mu r_1 + \nu r_2 = 1$. Let $f_1(x), f_2(x), \ldots f_k(x)$ be distinct irreducible polynomials of degree n and exponent e, where $e = r_1 r_2$. Let n_1 and n_2 be positive integers such that $kn = n_1 n_2$. Let $\alpha_u \in \mathbb{F}_{2^n}$ be a root of $f_u(x)$. Define $\beta_u = \alpha_u^{\nu r_2}$ and $\gamma_u = \alpha_u^{\mu r_1}$. Let $t_1, t_2, \ldots, t_k : \mathbb{F}_{2^n} \to \mathbb{F}_2$ be non-trivial $\mathbb{F}_2$-linear maps. For $1 \leq u \leq k$, define the binary $n_1 n_2 \times n$ matrix C_u as follows. The rows of C_u are indexed by pairs (i, j) where $0 \leq i < n_1$ and $0 \leq j < n_2$. The columns of C_u are indexed by integers v where $0 \leq v < n$. Then the entry of C_u in row (i, j) and column v is defined to be $t_u(\alpha_u^v \beta_u^i \gamma_u^j) \in \mathbb{F}_2$. Finally, define the $n_1 n_2 \times n_1 n_2$ matrix C to be the concatenation $C = (C_1 | C_2 | \cdots | C_k)$ of the matrices C_u.

Theorem 6: Let r_1 and r_2 be coprime positive integers. Let $f_1, f_2, \ldots f_k$ be distinct irreducible polynomials of degree n and exponent e, where $e = r_1 r_2$ and $n \geq 2$. Define $f(x) = \prod_{u=1}^{k} f_u(x)$. Let n_1 and n_2 be positive integers such that $n = n_1 n_2 / k$. Then the foldings of the non-zero sequences with characteristic polynomial dividing $f(x)$ form an $(r_1, r_2; n_1, n_2)$-PRAC if and only if $\det(C) \neq 0$, where C is the $n_1 n_2 \times n_1 n_2$ matrix defined above.

Proof: We use the notation defined in the statement of the theorem, and also in the paragraph above the statement.

Suppose that $\det(C) = 0$. Then there exists a non-zero (column) vector $\mathbf{d}$ such that $C\mathbf{d} = \mathbf{0}$. Write

$$\mathbf{d} = (d_{1,0}, d_{1,1}, \ldots, d_{1,n-1}, d_{2,0}, \ldots, d_{k,n-1})^T \tag{10}$$

for some $d_{u,v} \in \mathbb{F}_2$. Define $\delta_u \in \mathbb{F}_{2^n}$ by $\delta_u = \sum_{v=0}^{n-1} d_{u,v} \alpha_u^v$. Since $\mathbf{d}$ is a non-zero vector, not all the elements δ_u are zero. For all i and j with $0 \leq i < n_1$ and $0 \leq j < n_2$ we see that

$$\begin{aligned} \sum_{u=1}^{k} t_u(\delta_u \beta_u^i \gamma_u^j) &= \sum_{u=1}^{k} \sum_{v=0}^{n-1} t_u(d_{u,v} \alpha_u^v \beta_u^i \gamma_u^j) \\ &= \sum_{u=1}^{k} \sum_{v=0}^{n-1} d_{u,v} t_u(\alpha_u^v \beta_u^i \gamma_u^j) \\ &= 0, \end{aligned} \tag{11}$$

the last equality following by the definition of entries of the row of C indexed by (i, j) and the fact that $C\mathbf{d} = \mathbf{0}$.

Choose non-zero elements $\sigma_1, \sigma_2, \ldots, \sigma_k \in \mathbb{F}_{2^n}$ and $\sigma'_1, \sigma'_2, \ldots, \sigma'_k \in \mathbb{F}_{2^n}$ such that $\sigma_u + \sigma'_u = \delta_u$ for $1 \leq u \leq k$. Note that we can do this since $n \geq 2$. Moreover, note that

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$\sigma_u \neq \sigma'_u$ when δ_u is non-zero. Define binary sequences (s_i) and (s'_i) by

$$s_i = \sum_{u=1}^{k} t_u(\sigma_u \alpha_u^i) \text{ and } s'_i = \sum_{u=1}^{k} t_u(\sigma'_u \alpha_u^i). \tag{12}$$

We see that the characteristic polynomials of (s_i) and (s'_i) each divide $f(x)$, since $f(\alpha_u) = 0$ for all u. So the foldings of (s_i) and (s'_i) are shifts of elements in our array code. Note that $\sigma_u \neq \sigma'_u$ when δ_u is non-zero, so the sequences (s_i) and (s'_i) are not equal, and their foldings are therefore not the same. The sum of the foldings of (s_i) and (s'_i) is the folding of $(s_i + s'_i)$. But the folding of $(s_i + s'_i)$ has an $n_1 \times n_2$ all-zero window at its top-left corner, since for $0 \leq i < n_1 - 1$ and $0 \leq j < n_2 - 1$ we see that the (i, j) entry of the folding of $(s_i + s'_i)$ can be written as

$$\begin{aligned}
&\sum_{u=1}^{k} t_u(\sigma_u \beta_u^i \gamma_u^j) + \sum_{u=1}^{k} t_u(\sigma'_u \beta_u^i \gamma_u^j) \\
&= \sum_{u=1}^{k} t_u((\sigma_u + \sigma'_u)\beta_u^i \gamma_u^j) \\
&= \sum_{u=1}^{k} t_u(\delta_u \beta_u^i \gamma_u^j) \\
&= 0, \text{ by (11).}
\end{aligned}$$

The foldings of (s_i) and (s'_i) cannot form part of an $(r_1, r_2; n_1, n_2)$-PRAC, as they are distinct, but agree in their top left $n_1 \times n_2$ window. So we do not have a PRAC, as required.

Now suppose that $\det(C) \not= 0$. We will now show that the foldings of the non-zero sequences with characteristic polynomial dividing $f(x)$ form an $(r_1, r_2; n_1, n_2)$-PRAC. Let (s_1) and (s'_i) be two such sequences, and suppose that their foldings agree in an $n_1 \times n_2$ window. Without loss of generality, by taking suitable shifts of our sequences, we may assume that the window is at the top left of the array. To show we have a PRAC, it suffices to show that $(s_i) = (s'_i)$.

Since (s_i) and (s'_i) have characteristic polynomial dividing $f(x)$, we may write them in the form (12) for some elements $\sigma_u, \sigma'_u \in \mathbb{F}_{2^n}$. Defining $\delta_u = \sigma_u + \sigma'_u$ for all u, we have

$$s_i + s'_i = \sum_{u=1}^{k} t_u(\sigma_u \alpha_u^i) + \sum_{u=1}^{k} t_u(\sigma'_u \alpha_u^i) = \sum_{u=1}^{k} t_u(\delta_u \alpha_u^i).$$

We may write $\delta_u = \sum_{v=0}^{n-1} d_{u,v} \alpha_u^v$ for some elements $d_{u,v} \in \mathbb{F}_2$. The top left $n_1 \times n_2$ window of the folding of the sequence $(s_i + s'_i)$ is all zero, and hence for all $0 \leq i < n_1$ and $0 \leq j < n_2$

$$\begin{aligned}
0 &= \sum_{u=1}^{k} t_u(\delta_u \beta_u^i \gamma_u^j) \\
&= \sum_{u=1}^{k} \sum_{v=0}^{n-1} t_u(d_{u,v} \alpha_u^v \beta_u^i \gamma_u^j) \\
&= \sum_{u=1}^{k} \sum_{v=0}^{n-1} d_{u,v} t_u(\alpha_u^v \beta_u^i \gamma_u^j).
\end{aligned}$$

Defining $\mathbf{d}$ using (10), we see that the (i, j) entry of $C\mathbf{d}$ is 0 for all $0 \leq i < n_1$ and $0 \leq j < n_2$, and so $\mathbf{d}$ lies in the null space of C. As $\det(C) \neq 0$, the matrix C has trivial null space and so $\mathbf{d} = 0$. Thus $d_{u,v} = 0$ for all u and v, which implies that $\delta_u = 0$ for $1 \leq u \leq k$. Since $\sigma_u + \sigma'_u = \delta_u = 0$, we see that $\sigma_u = \sigma'_u$ for $1 \leq u \leq k$, and so $(s_i) = (s'_i)$ as required. ∎

Note that for any fixed u with $1 \leq u \leq k$, the elements α_u^v for $0 \leq v < n$ are a basis for $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. These elements occur in the definition of the matrix C, but they can be replaced by any other basis for $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. The proof of the theorem is essentially the same if this change is made.

B. A Dual Condition

The following corollary (giving a 'dual' condition to that in Theorem 6 in the special case when $k = 1$) is very close in spirit to Theorem 5.

Corollary 15: Let r_1 and r_2 be coprime positive integers. Let μ and ν be integers such that $\mu r_1 + \nu r_2 = 1$. Let $f(x)$ be an irreducible polynomial of degree n and exponent e, where $e = r_1 r_2$ and $n \geq 2$. Let n_1 and n_2 be positive integers so that $n = n_1 n_2$. Let $\alpha \in \mathbb{F}_{2^n}$ be a root of $f(x)$. Define $\beta = \alpha^{\nu r_2}$ and $\gamma = \alpha^{\mu r_1}$. Regard the elements of $\mathbb{F}_{2^n}$ as binary vectors of length n. Then the foldings of the non-zero sequences with characteristic polynomial $f(x)$ form an $(r_1, r_2; n_1, n_2)$-PRAC if and only if the following $n_1 n_2 = n$ vectors in $\mathbb{F}_2^n$ are linearly independent:

$$\beta^i \gamma^j \text{ where } 0 \leq i < n_1 \text{ and } 1 \leq j < n_2.$$

Proof: Let C be the $n \times n$ matrix whose rows are indexed by pairs (i, j) with $0 \leq i < n_1$ and $0 \leq j < n_2$, columns indexed by integers v with $0 \leq v < n$, where the entry in row (i, j) and column v is equal to

$$c_{(i,j),v} = \mathrm{Tr}(\alpha^v \beta^i \gamma^j).$$

By Theorem 6 (in the case when $t_u = \mathrm{Tr}$ for $1 \leq u \leq k$), our folding fail to form a PRAC if and only if $\det(C) = 0$.

Let $\mathbf{d} = (d_0, d_1, \ldots, d_{n-1})^T$, with $d_v \in \mathbb{F}_2$ for $0 \leq v < n$, be a column vector. Let $\delta = \sum_{v=0}^{n-1} d_v \alpha^v$ be the corresponding element of $\mathbb{F}_{2^n}$. For any pair (i, j) with $0 \leq i < n_1$ and $0 \leq j < n_2$, we have that

$$\begin{aligned}
\sum_{v=0}^{n-1} c_{(i,j),v} d_v &= \sum_{v=0}^{n-1} d_v \mathrm{Tr}(\alpha^v \beta^i \gamma^j) \\
&= \mathrm{Tr}\left(\left(\sum_{v=0}^{n-1} d_v \alpha^v\right) \beta^i \gamma^j\right) \\
&= \mathrm{Tr}(\delta \beta^i \gamma^j).
\end{aligned}$$

Thus $\mathbf{d}$ lies in the (column) null space of C if and only if

$$\mathrm{Tr}(\delta \beta^i \gamma^j) = 0 \text{ for } 0 \leq i < n_1 \text{ and } 0 \leq j < n_2. \tag{13}$$

Define the $\mathbb{F}_2$-linear map $t : \mathbb{F}_{2^n} \to \mathbb{F}_2$ by $t(z) = \mathrm{Tr}(\delta z)$ for all $z \in \mathbb{F}_{2^n}$. Suppose the elements $\beta^i \gamma^j$ are linearly independent, so they form a basis for $\mathbb{F}_{2^n}$. The conditions (13) then imply that t is the trivial map. Hence, by [20, Theorem 2.24], we have $\delta = 0$. So the null space of C is trivial and therefore $\det(C) \neq 0$. Hence our folding form a PRAC.

If the elements $\beta^i \gamma^j$ are linearly dependent, the n linear constraints (13) are dependent, and so there exist non-zero solutions δ to these constraints. Thus the null space of C is non-trivial, so $\det(C) = 0$, and hence our folding does not form a PRAC. ∎

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X. Conclusion and Future Research

Pseudo-random arrays and pseudo-random array codes are constructed by folding sequences generated by irreducible polynomials, or by reducible polynomials that are a product of distinct irreducible polynomials of the same degree and the same exponent. The structures constructed by folding are examined. Necessary and sufficient conditions to verify whether the array or the array code is a PRA or a PRAC, respectively, are given. Several techniques that require simple computations for divisibility of integers to verify if some of these structures are PRAs or PRACs are proved. One set of conditions that generalizes the conditions given by MacWilliams and Sloane [22] for generating one pseudo-random array is shown to yield pseudo-random arrays not known before. This set of conditions was generalized for pseudo-random array codes. Another completely different set of conditions, with apparently a different construction for pseudo-random array codes that can be also generated by folding, is also presented. Finally, the codes constructed by the different constructions are compared and two hierarchies implied by the constructions are discussed.

There are many other interesting problems that arise from our discussion. Some of them are as follows:

1) We did not consider constructions of $(r_1, r_2; n_1, n_2)$-SDBACs which are not PRACs and might have parameters not obtained by PRACs. It is not difficult to construct them for $n_2 = 2$. Beyond $n_2 = 2$ it is an intriguing problem to construct such array codes which are not PRACs. This is a promising direction for future research.
2) Some of the constructed pseudo-random array codes have large minimum distance. This is mentioned with an example in [9]. An analysis of the minimum distance of these codes is an interesting question. These codes are linear codes, and it is interesting to compare their parameters with other linear array codes.
3) Experimental results show that there exist more parameters for which folded sequences generated by an irreducible polynomial produces a PRA or a PRAC beyond Theorem 2 and Theorem 5. Examples of such pseudo-random array codes were presented in Section V. We would like to see a comprehensive solution for this case, i.e., for which parameters and for which M-sequences, does folding yield pseudo-random arrays. Similarly, for which parameters and for which sequences generated by an INP polynomial does folding yield pseudo-random array codes?
4) Experiments show that folding sequences generated by reducible polynomials whose factors are distinct irreducible polynomials with the same degree and the same exponent generates a pseudo-random array. Only a few cases are covered by Theorem 4 and Corollary 8. We would like to see a comprehensive treatment of folding for such sequences.
5) We would like to see a proof of, or a counterexample for, Conjecture 1 in Section VIII.

Acknowledgment

The authors thank Ronny Roth for pointing out Theorem 8.67 in [20]. They also thank the anonymous reviewers for their careful reading which contributes to the presentation of the results in the article.

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